



February 2025

Glacial hazards

Risk assessment and mitigation

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fidelsteiner.github.io/#teaching

Topics

Terminology: Hazards. Risks. Mitigation.

Global ice and exposure to hazards

Types of glacial hazards

The global politics of 'glacial risks' - excursion to Greenland

Hazard Assessment: Mapping and Modelling

Monitoring and risk mitigation

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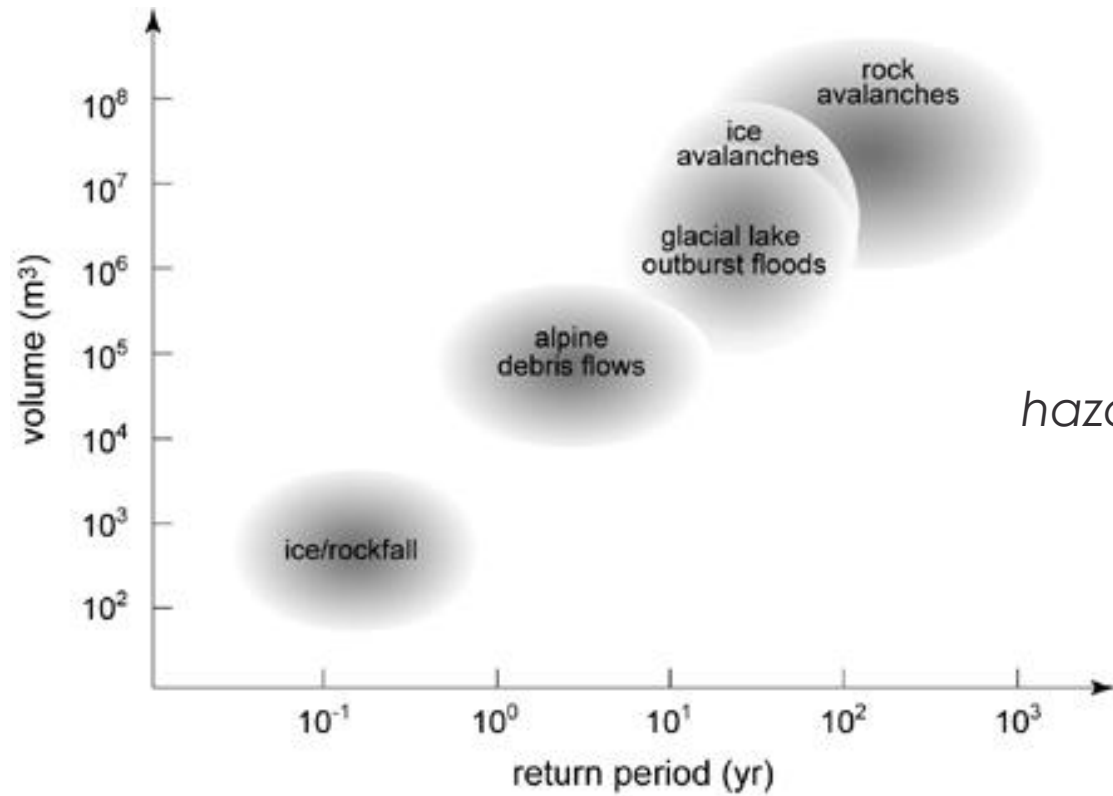
Glacial hazards vs cryosphere hazards

Cryosphere = glaciers – snow – permafrost – sea/lake ice

Other cryosphere hazards are...

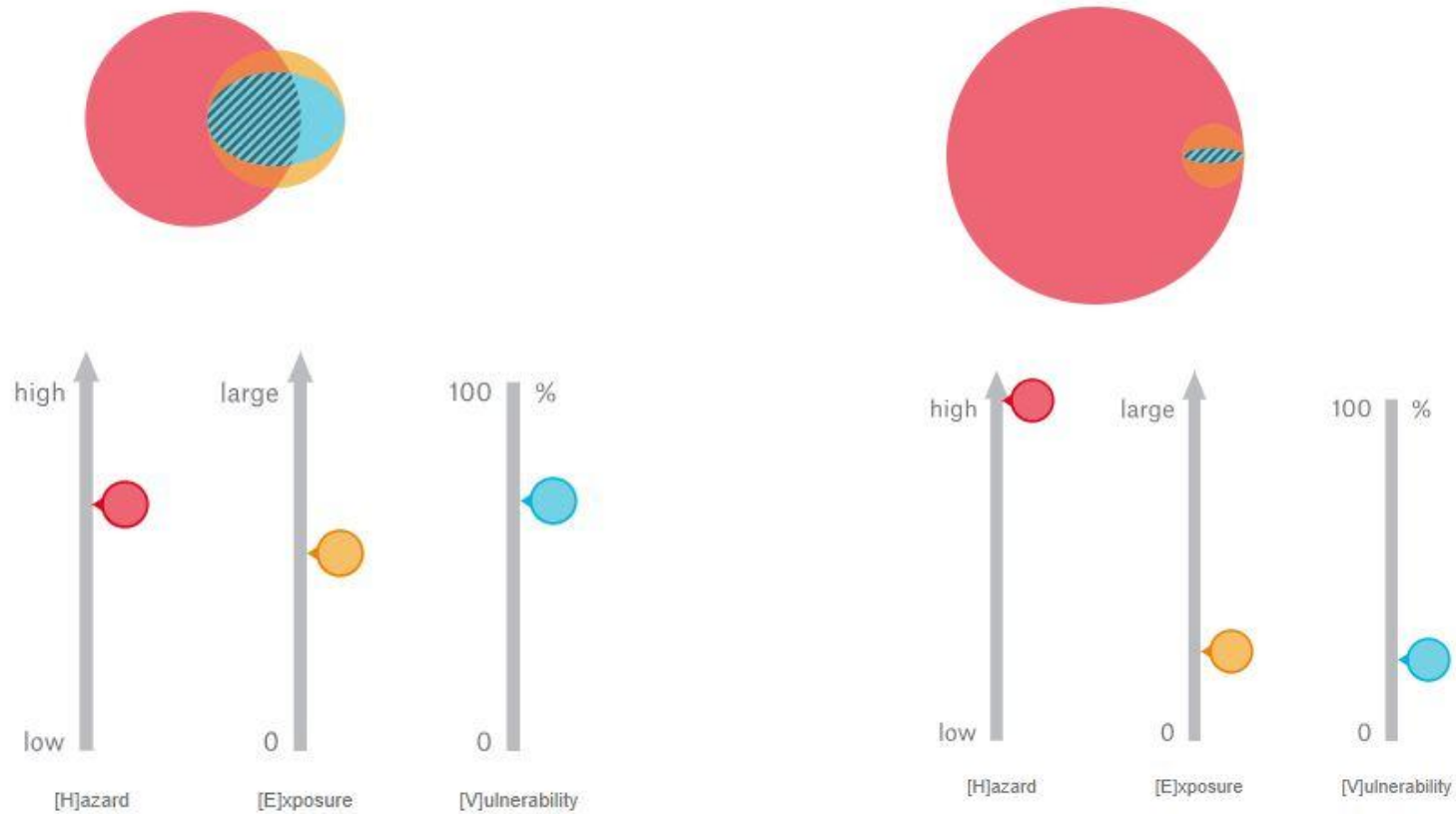
- snow avalanches
- snow droughts
- permafrost change induced slope instabilities
- change of sea ice breakup

Return periods vs. volumes



hazard = f(probability, intensity)

risk = ***hazard*** * ***exposure*** * ***vulnerability***
[*adaptive capacity *insurance penetration]



Mitigation and adaptation

Mitigation – reducing risk from the occurrence of any undesirable event or minimize the degree of damage

Adaptation – prepare for damages that are to be expected (and possibly can't be avoided or mitigated)

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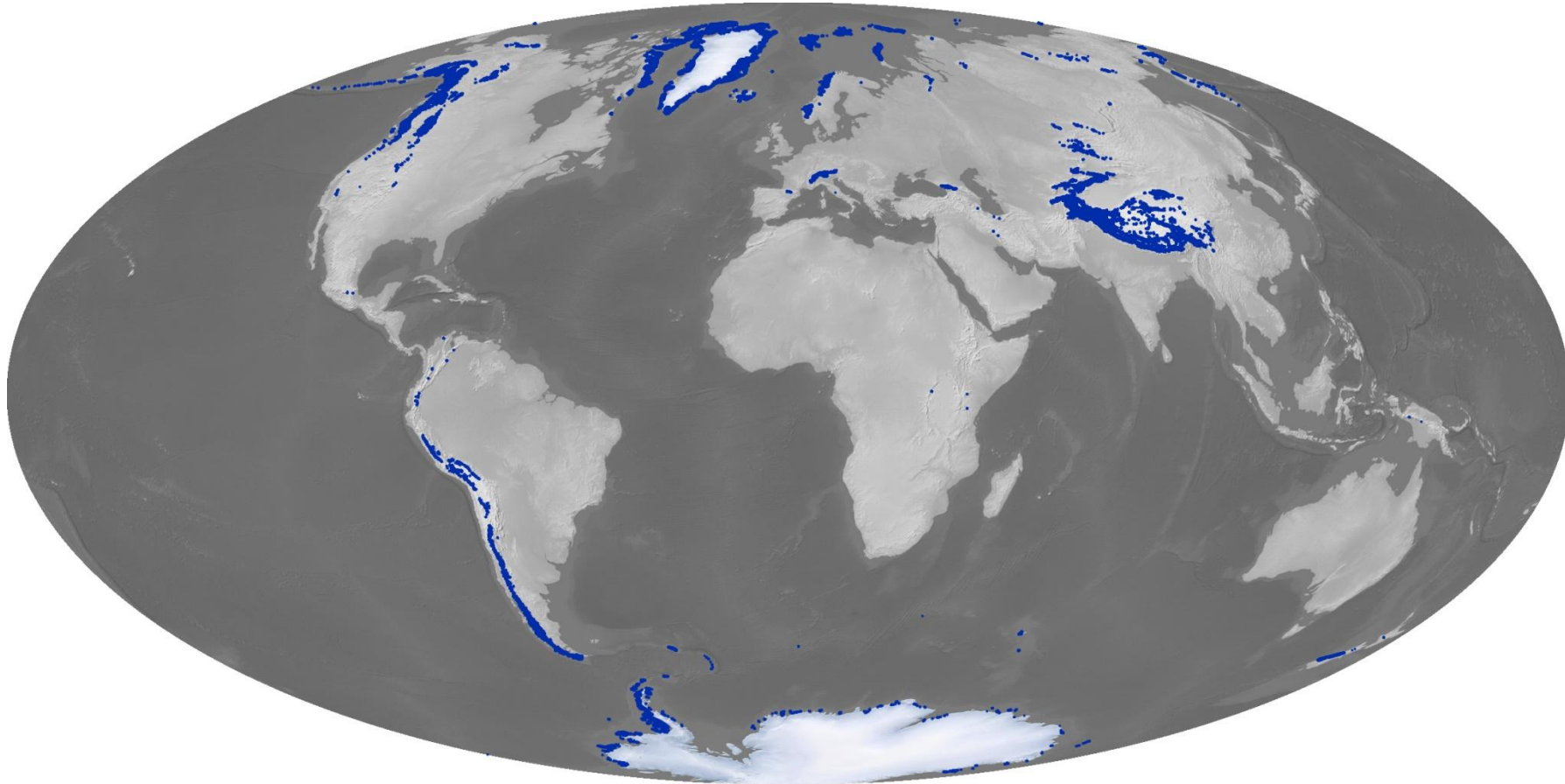
Monitoring and risk mitigation

Glaciers globally

Total Area:

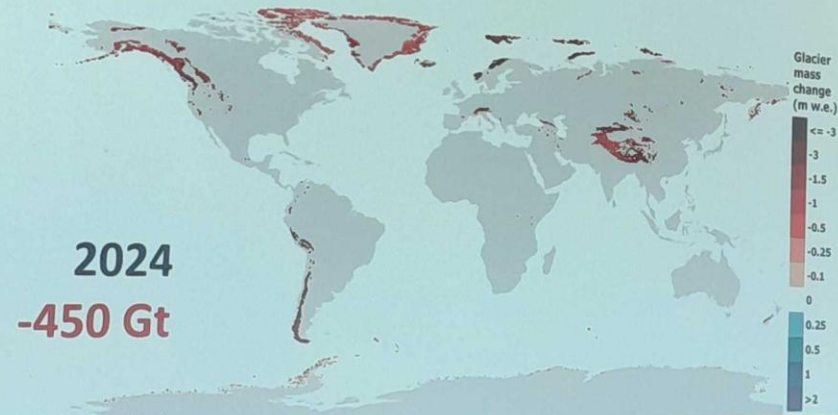
RGI: 734 933 km²

WGI: 747 688 km²



R. Simmon, NASA; based on RGI4
(2013)

How fast are glaciers melting?



2024
-450 Gt

Dussaillant et al. (2025, in review)

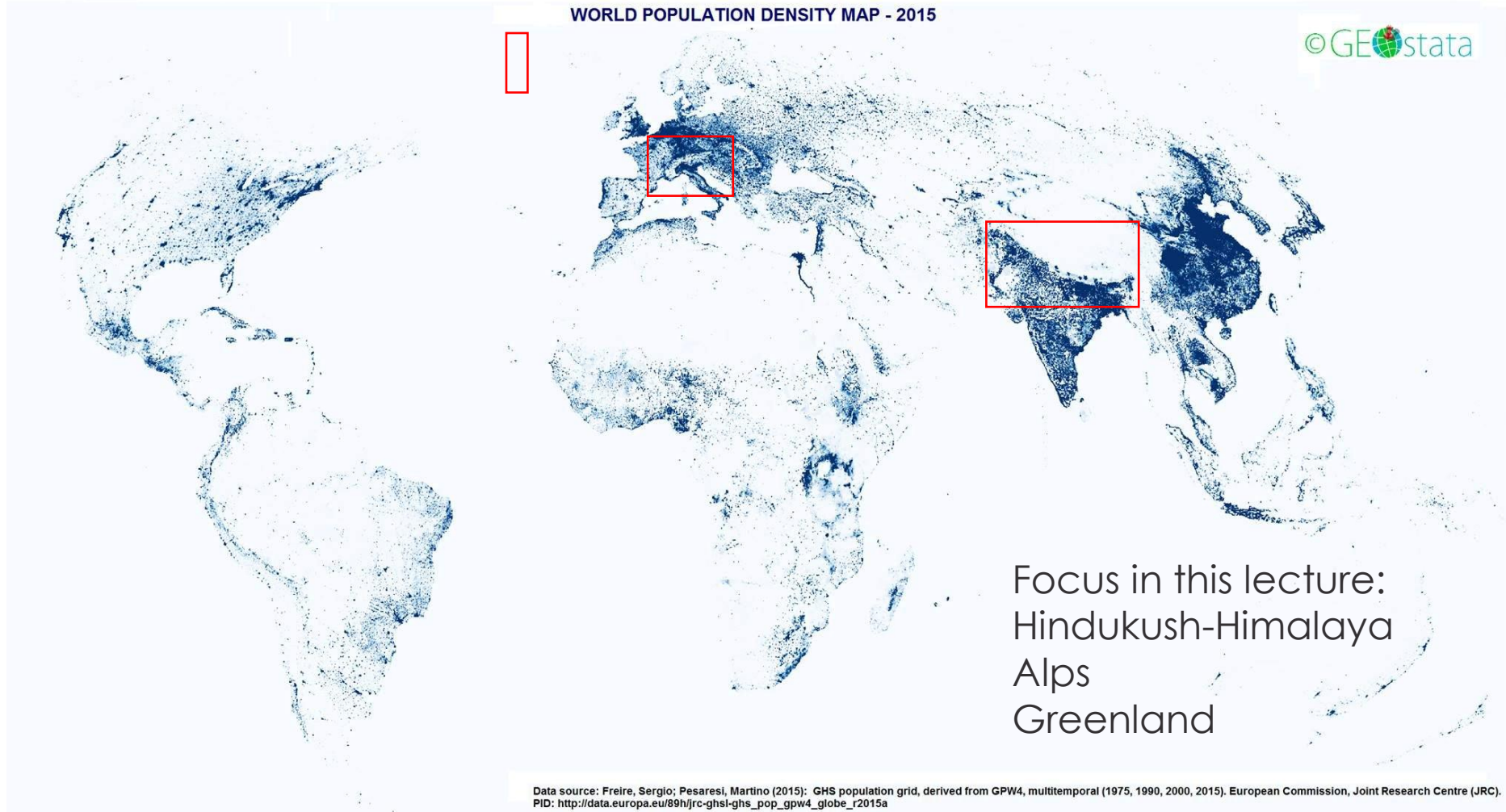
We can by now document
change rapidly

1928's initiative



... but it also pays to
think ahead

WORLD POPULATION DENSITY MAP - 2015



Focus in this lecture:
Hindukush-Himalaya
Alps
Greenland

Data source: Freire, Sergio; Pesaresi, Martino (2015): GHS population grid, derived from GPW4, multitemporal (1975, 1990, 2000, 2015). European Commission, Joint Research Centre (JRC).
PID: http://data.europa.eu/89h/jrc-ghsl-ghs_pop_gpw4_globe_r2015a

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Glacier related hazards

glacial lake outburst floods (+ resulting events)

ice break offs (+ resulting events) / collapsing glaciers

glacier length variations

destabilization of frozen soils/slopes

... what appears when the ice is gone?

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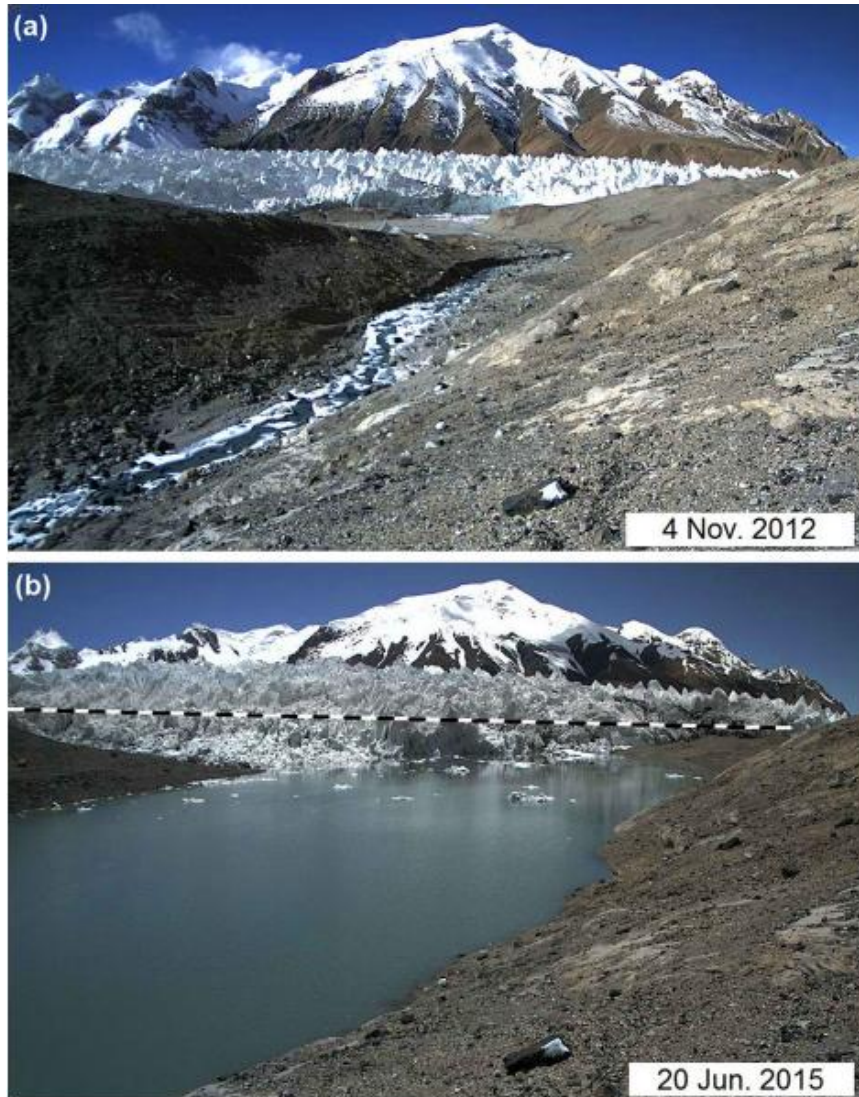
Glacial Lake Outburst Floods (1)

British Columbia
Canada



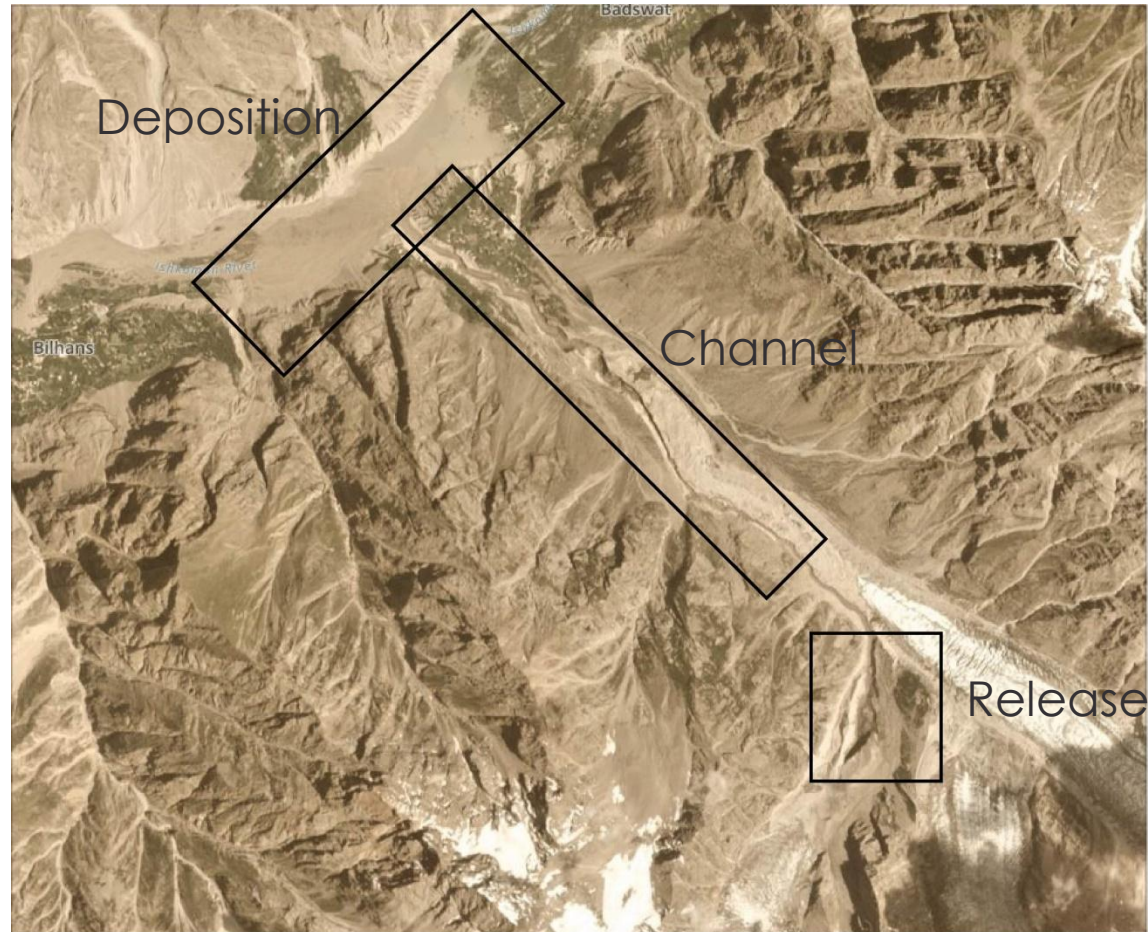
Fig. 12. Aerial photographs of Nostetuko Lake before (left, BC79069-190) and after (right, BC86048-147) the outburst flood of July 19, 1983. Failure occurred when toe of Cumberland Glacier (arrow) collapsed into the lake and generated waves that overtopped the moraine.

Glacial Lake Outburst Floods (2)

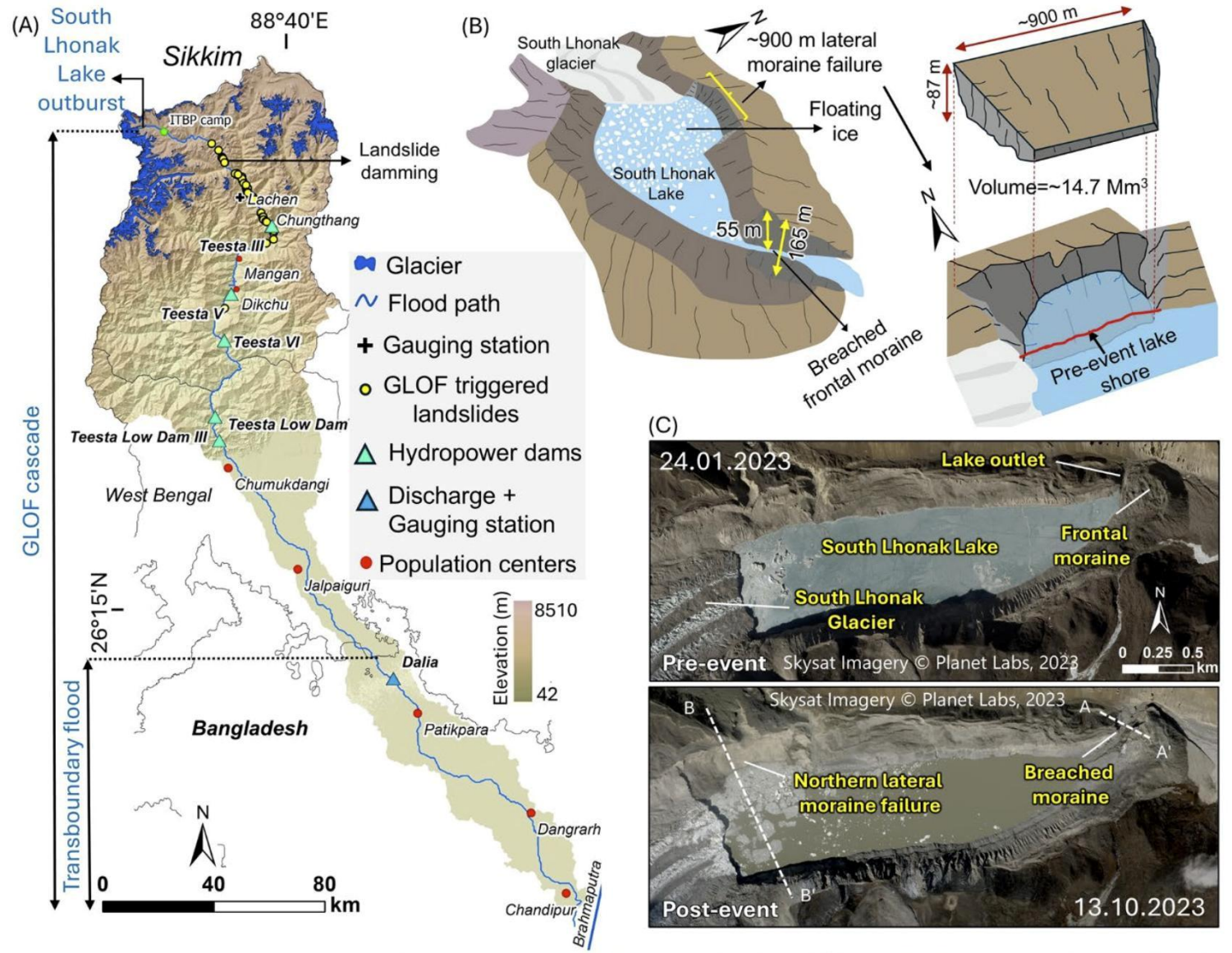


Kyagar Lake, Xinjiang, China, Geopraevent

Glacial Lake Outburst Floods (3)

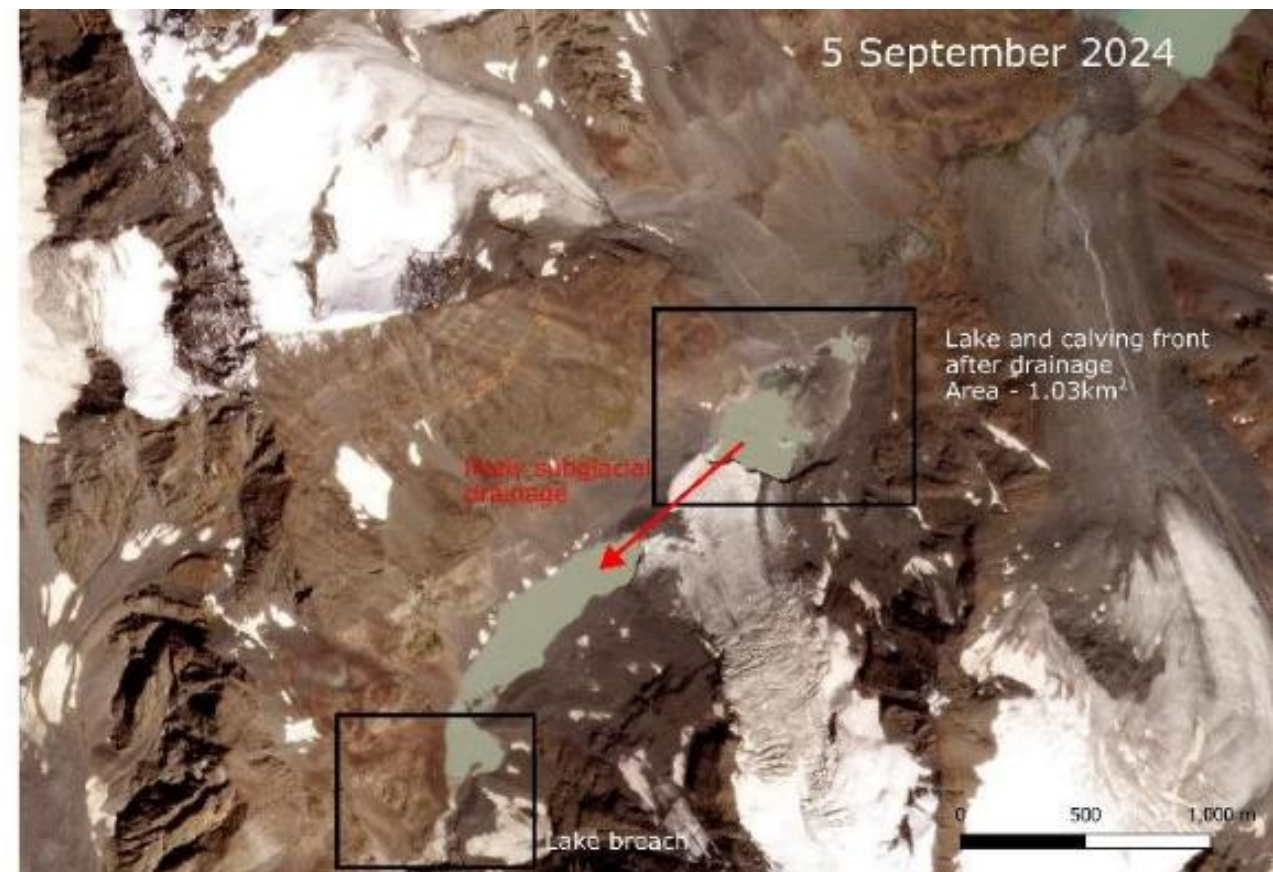
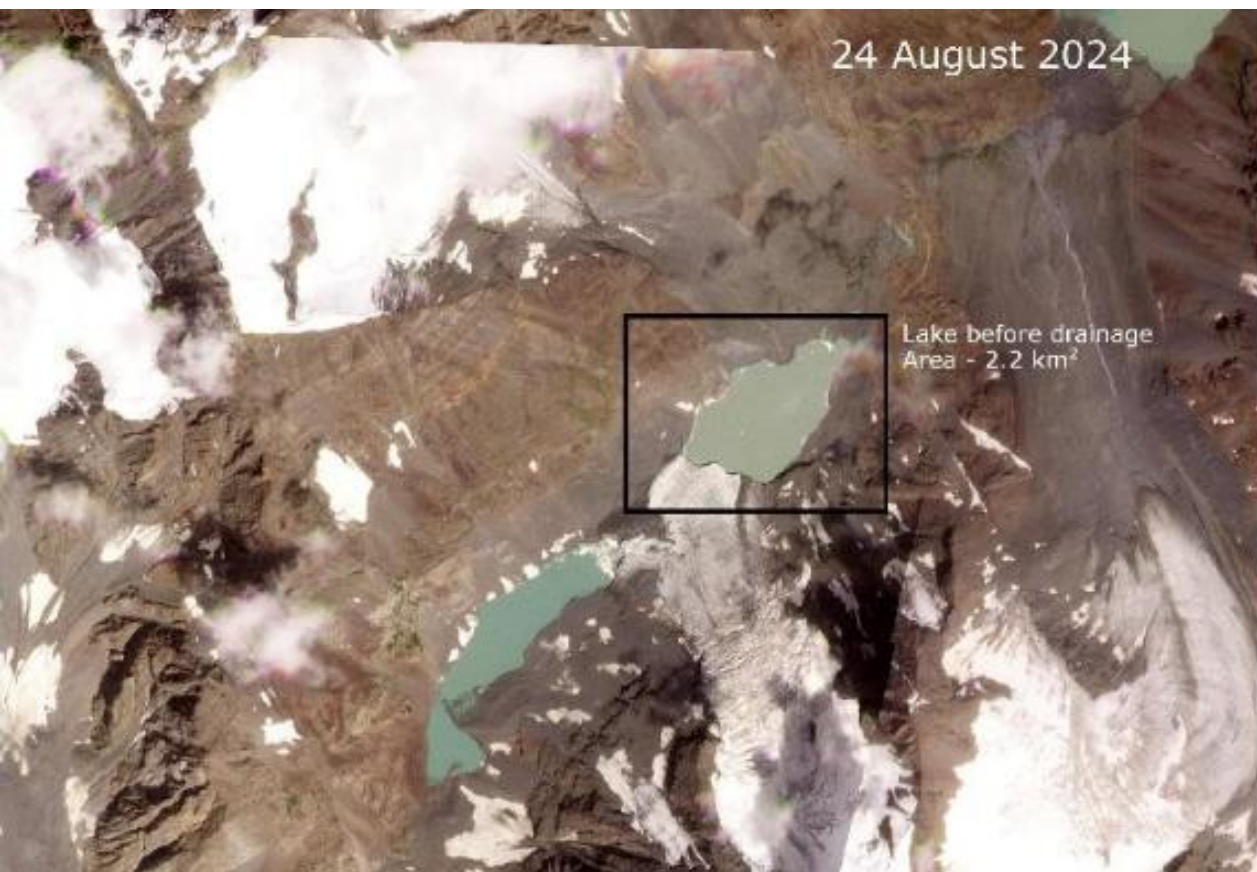


Helicopter Image/Pakistan Army; Sat image Planet

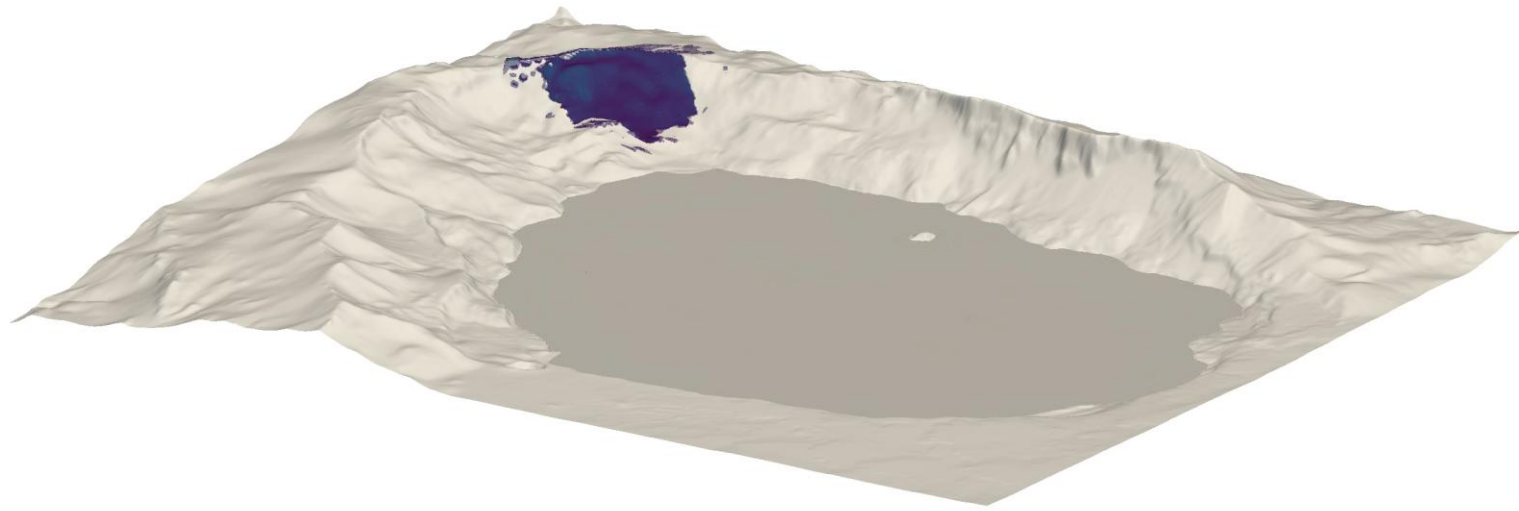


Sattar et al. 2025;

<https://www.carbonbrief.org/catastrophic-2023-lake-outburst-in-india-driven-by-glacial-melt-and-permafrost-thaw/>



Impact waves



Glacial Lake Outburst Floods - Causes

Mass movement into the terminal lake causing overtopping (destabilizing glacier/rock slopes)

Failure of a moraine dam (melting of ice cored moraine)

Surging glaciers damming a river (surges/melt season)

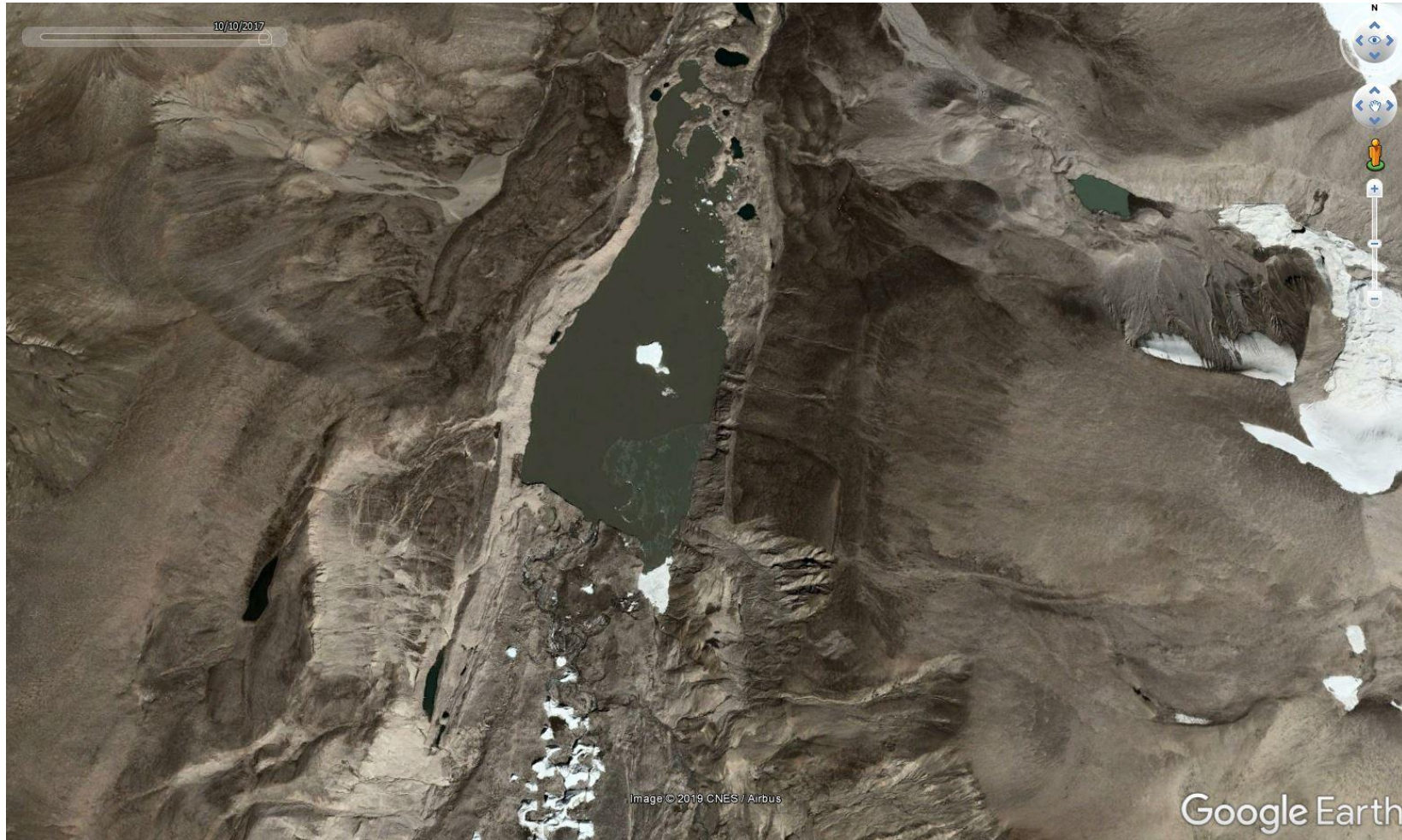
Failure of an englacial lake (internal melt)

Intense melt/rainfall

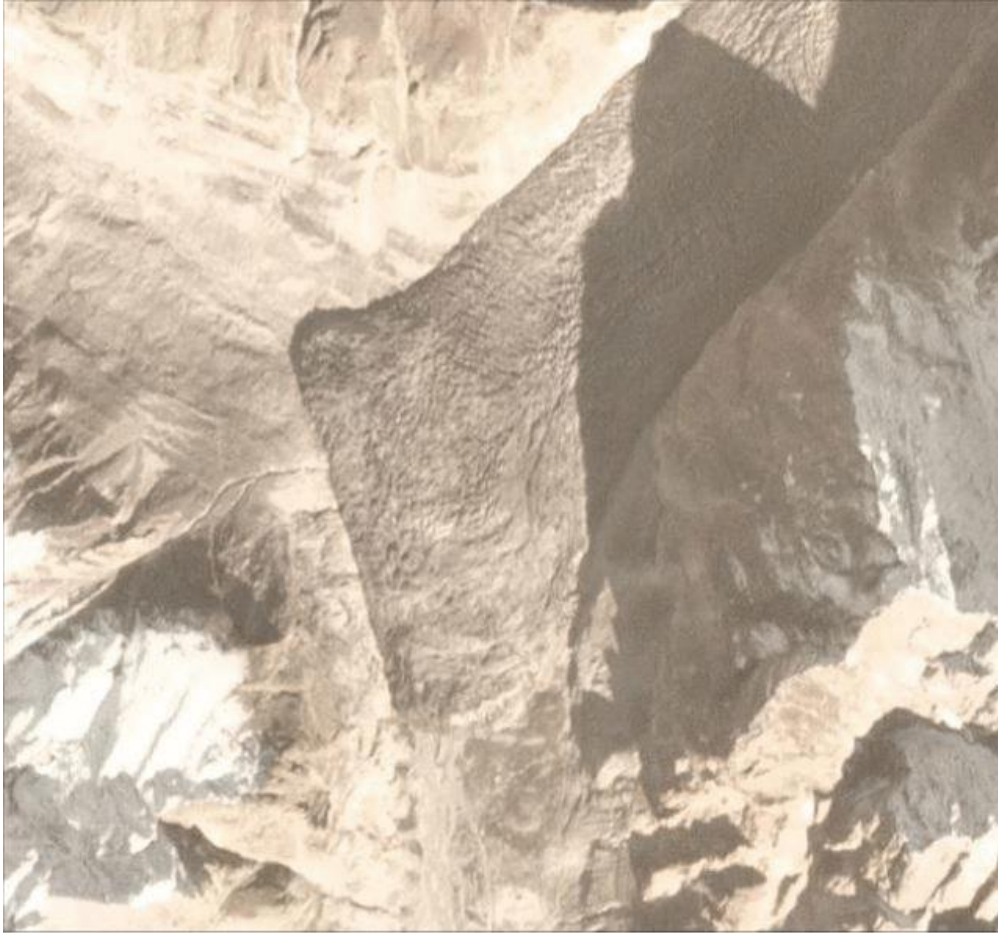
Growth of terminal lakes (2012)



Growth of terminal lakes (2017)



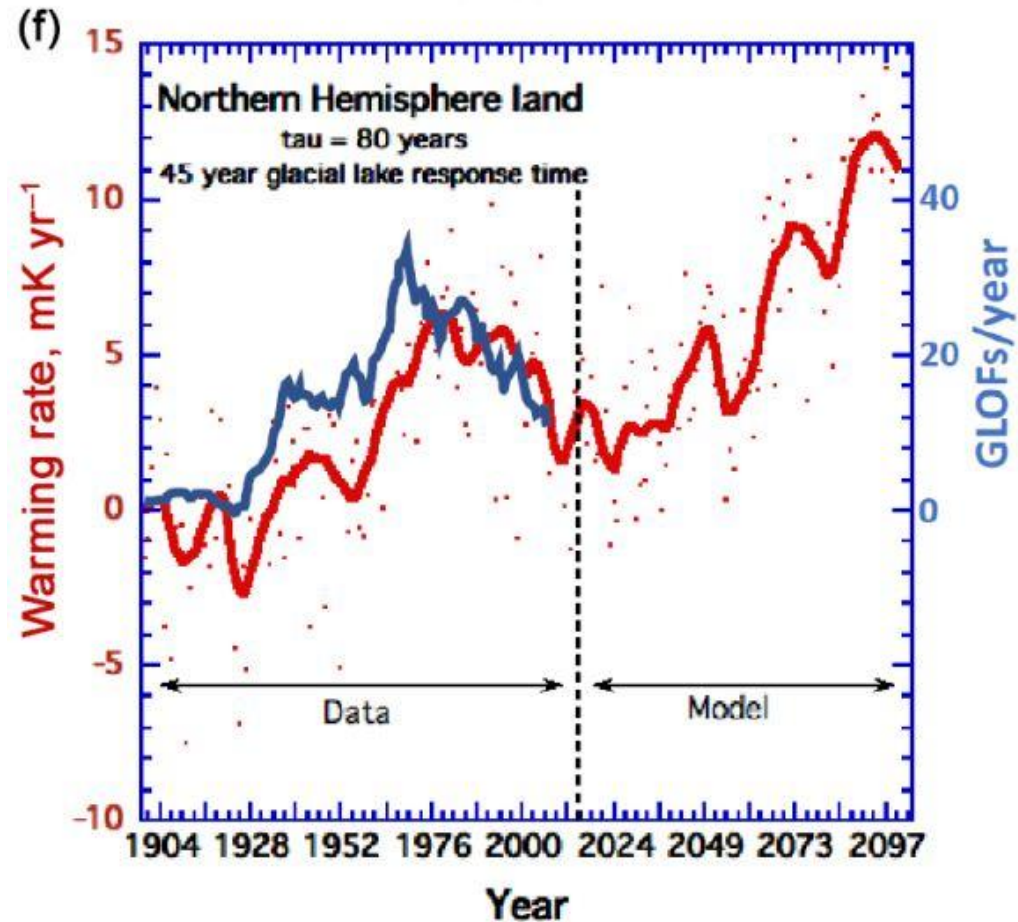
Surging glaciers



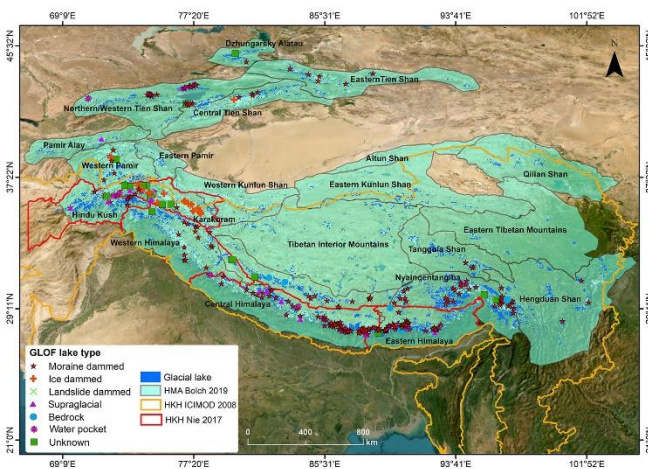
- Rapidly advancing glaciers block rivers
- Potentially hazardous lakes
- Interference with water supply
- Karakoram, Svalbard, Andes, Alaska

Glacier Lake Outburst Floods - Causes

- Link to climate change?
 - Seems very plausible and some indications
 - Difficult to proof with very little data
 - Data bias towards more observations in recent years



Glacial Lake Outburst Floods - Causes

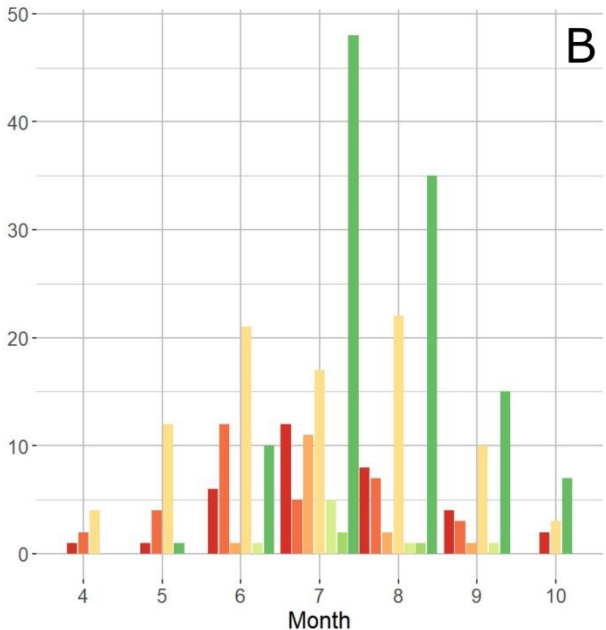
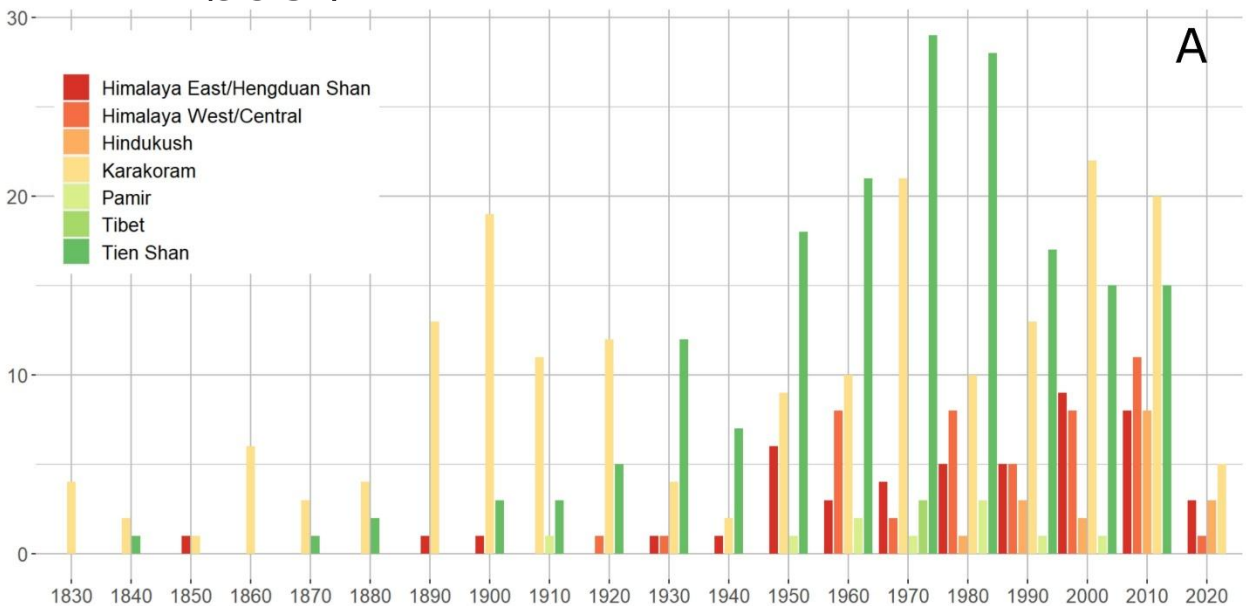


In High Mountain Asia ca 1000 people died in the last 100 years

Low frequency – high impact events

(Avalanches: 3000 people died, high frequency, medium impact;

Debris flows: preliminary 8000 people died, high frequency, varied impact)



Glacial Lake Outburst Floods - *Results*

Lake outbursts from moraine dammed lakes are stagnating, possibly increasing (Veh et al., 2022)

Lake outbursts from ice dammed lakes are decreasing (Veh et al., 2023)

Lake outbursts are decreasing in size, although lakes are increasing in size (Veh et al. 2025)

Outbursts from inside the ice (*waterpockets*) are an uncertain thing we need to investigate further

Gla

Lake o
2022)

Lake o

Outbur
further

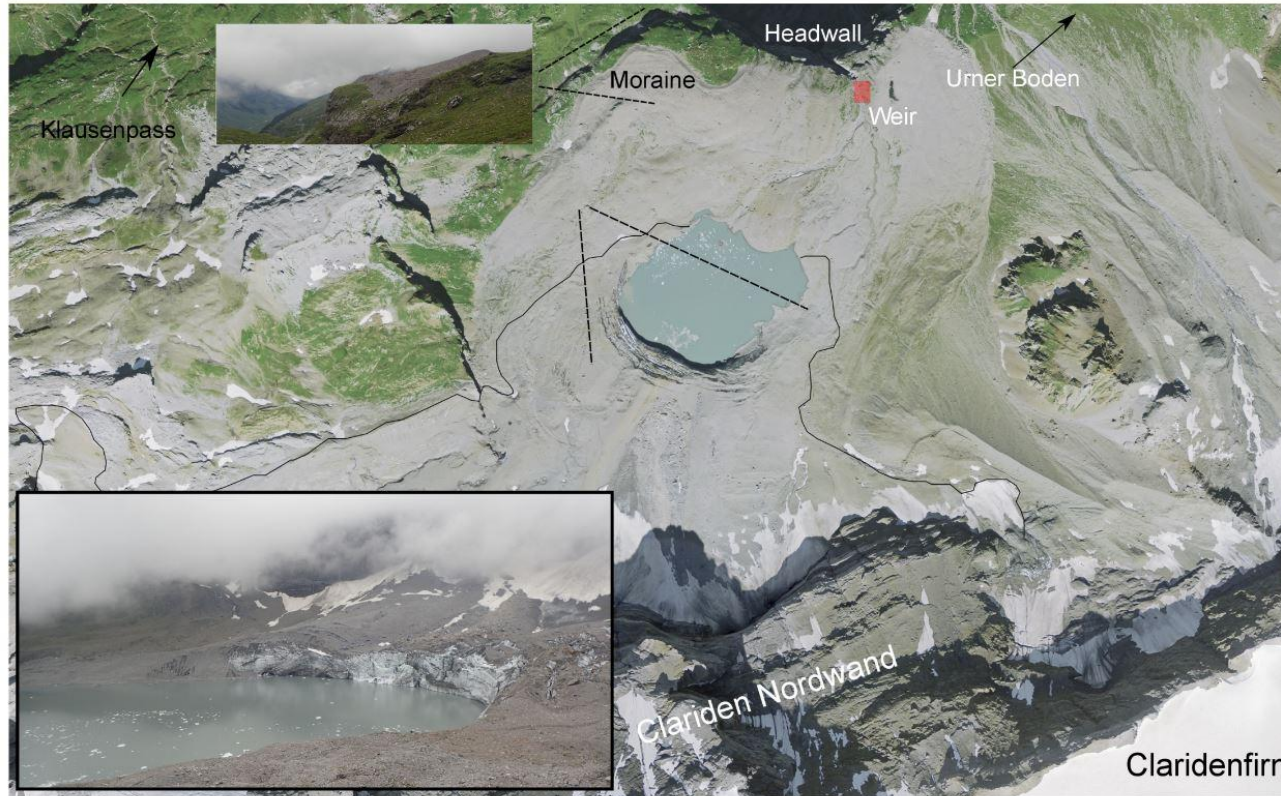
et al.,

igate



Tête Rousse (Mont Blanc Massif) in 1892; Vincent et al., 2017

Glacial Lake Outburst Floods - *Results*

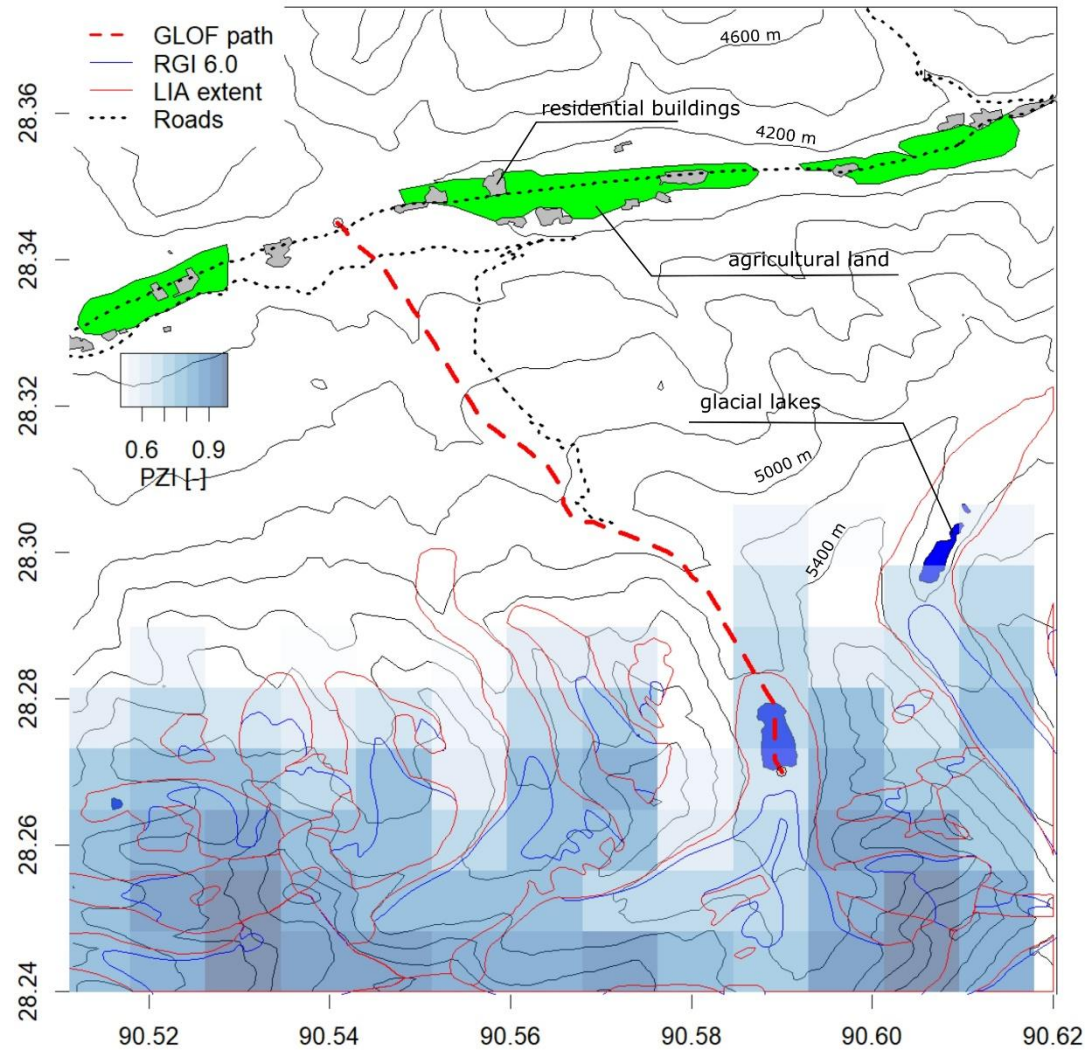


- Relatively small lake in Switzerland
- Rapid growth due to glacier retreat
- Potential threat from ice break offs into the lake
- Huge damage potential due to expensive hydropower plant below

Figure 1: Griessee in an orthophoto from 2013 (main frame, swisstopo). The calving front as well as the frontal moraine of the glacier are shown in insets taken in August 2016 by the authors.

Steiner et al, 2016

Glacial Lake Outburst Floods - *Results*



Shrestha et al, 2023

Glacier related hazards

glacier lake outburst floods (+ resulting events)

ice break offs (+ resulting events) / collapsing glaciers

glacier length variations

destabilization of frozen soils/slopes

... what appears when the ice is gone?

Ice break offs + collapsing glaciers

- Increasingly dangerous with receding glacierettes/hanging glaciers
- Added danger in seismically active regions

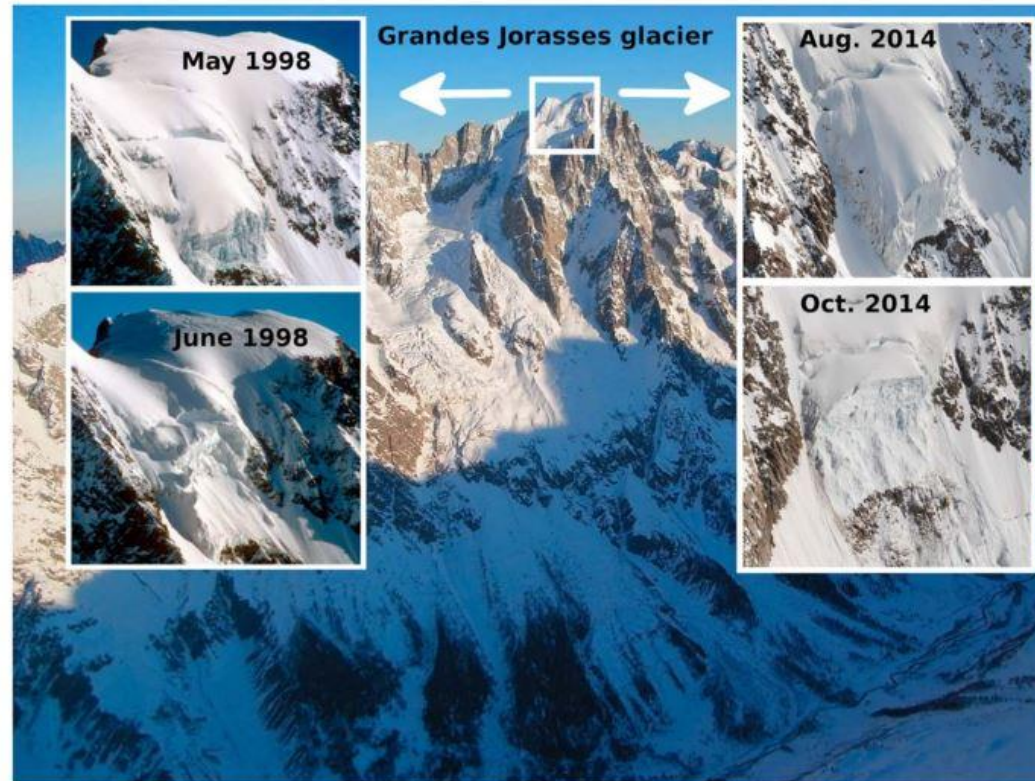


Figure 7. South side of the Grandes Jorasses and the Italian Val Ferret. (left inset) Evolution of the hanging glacier from May to June 1998. (right inset) Evolution of the hanging glacier from August to October 2014.

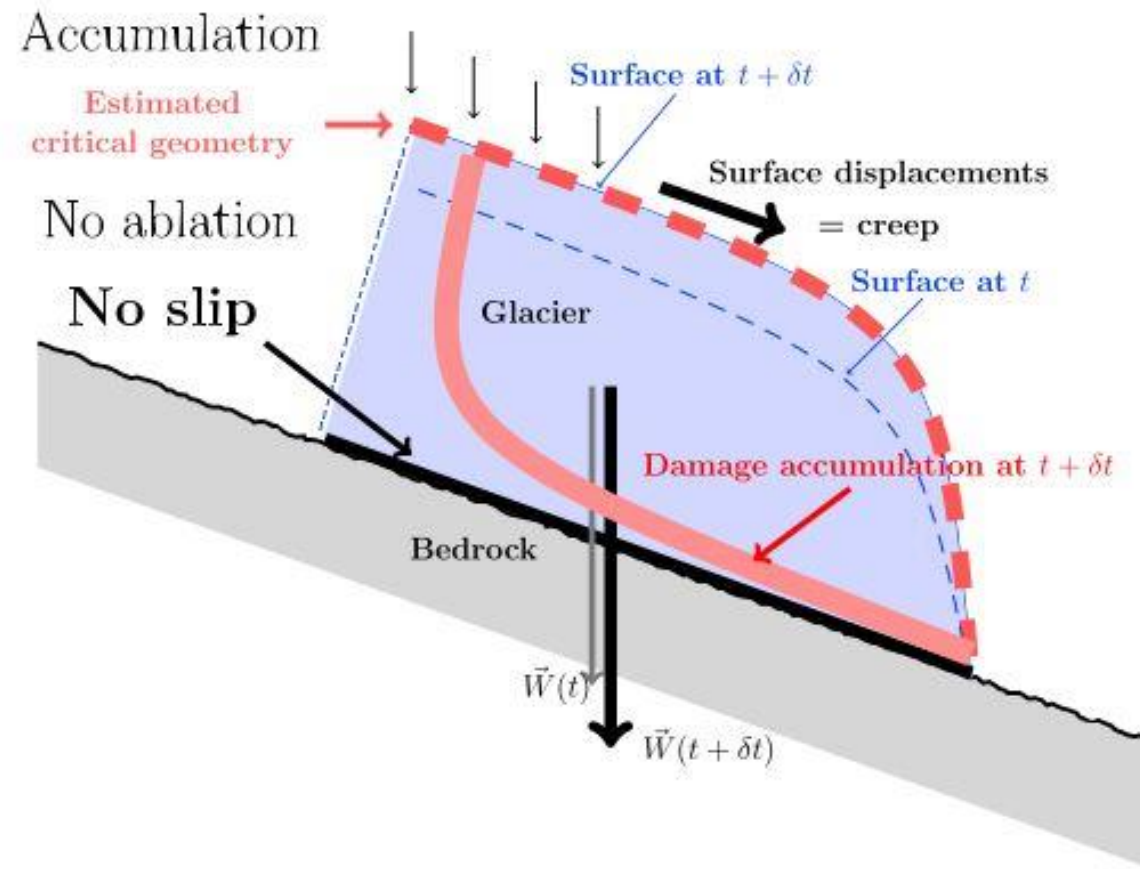
Ice break offs + collapsing glaciers





Causes – *cold based: damage propagation*

- 3 glacier types
 - cold based
 - warm based
 - polythermal



Causes – *transition from cold to warm based*

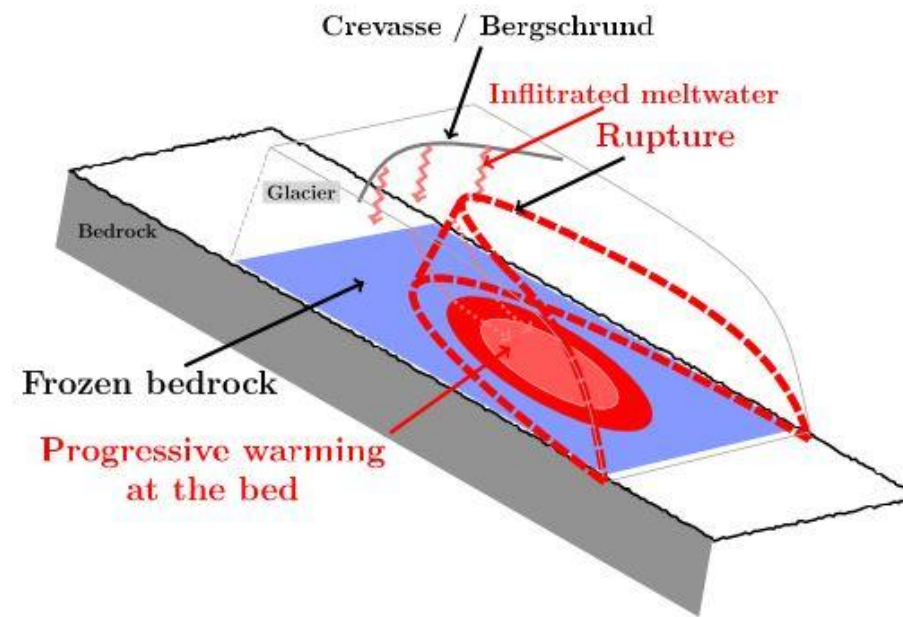


Figure 12. Schematic evolution of the instability initiated by a rapid localized warming at the ice-bedrock interface.

Causes – *warm based; hydraulic slip*

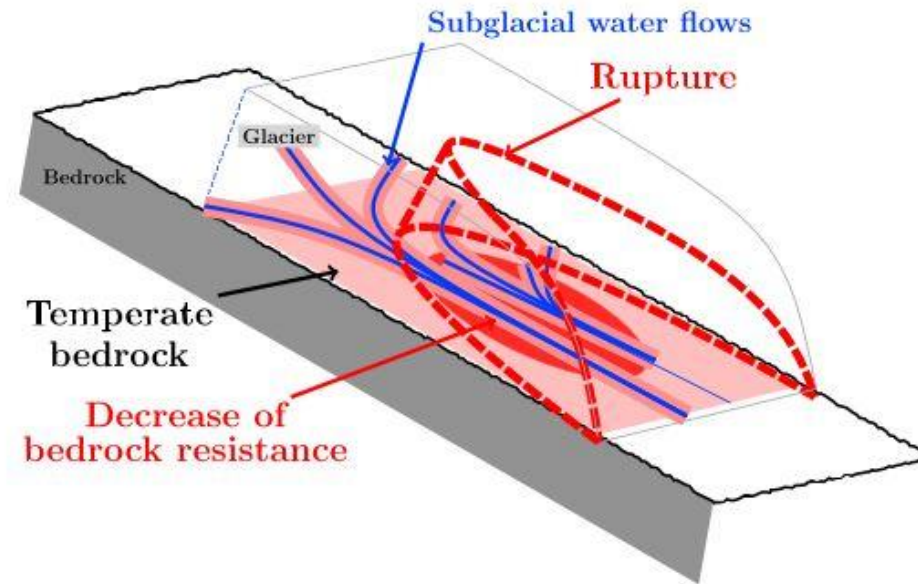


Figure 16. Schematic evolution of the instability for the case of a balanced glacier with a decreasing basal resistance due to increasing subglacial water pressure.

Results – disasters in remote locations (Uttarakhand, India, 7 February 2021)

At 10:30 local time a major debris flow was witnessed and shared on social media, hitting a hydropower plant under construction

By 19:00 local time satellite imagery of the event had become available that allowed us to document the reason for the event

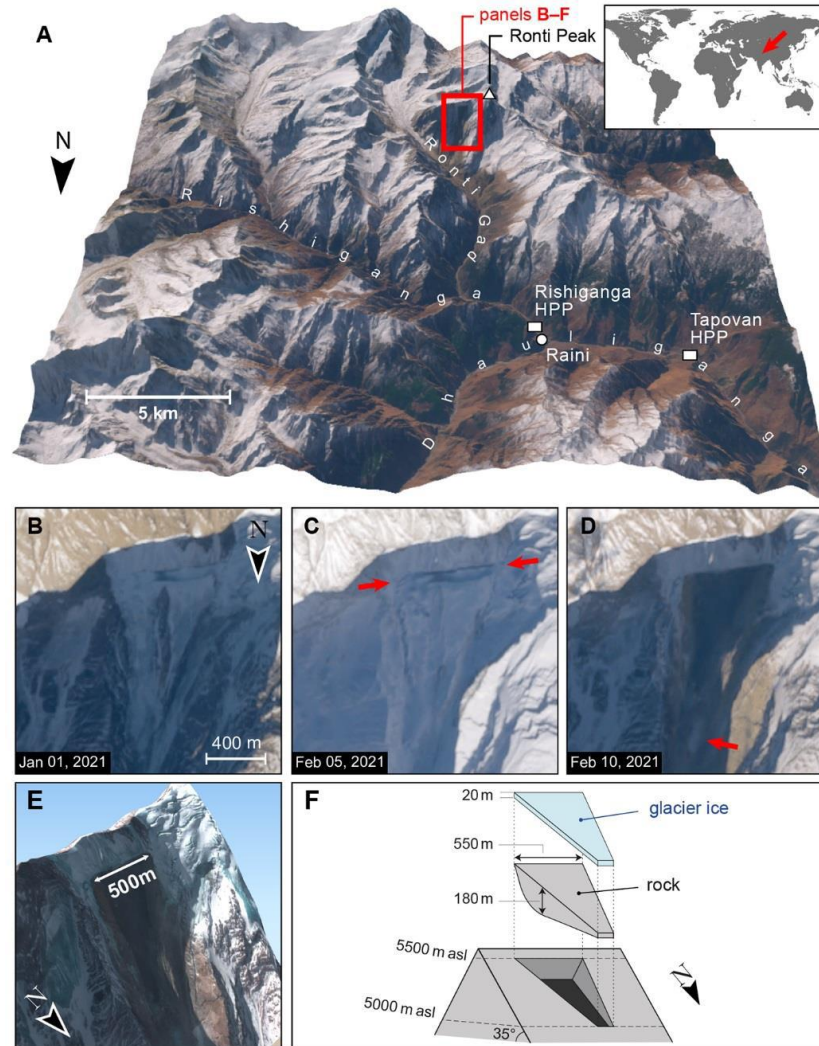
>200 people dead

Results – Uttarakhand rock fall/debris flow

rockfall mixed with ice

mobilization of deposited debris

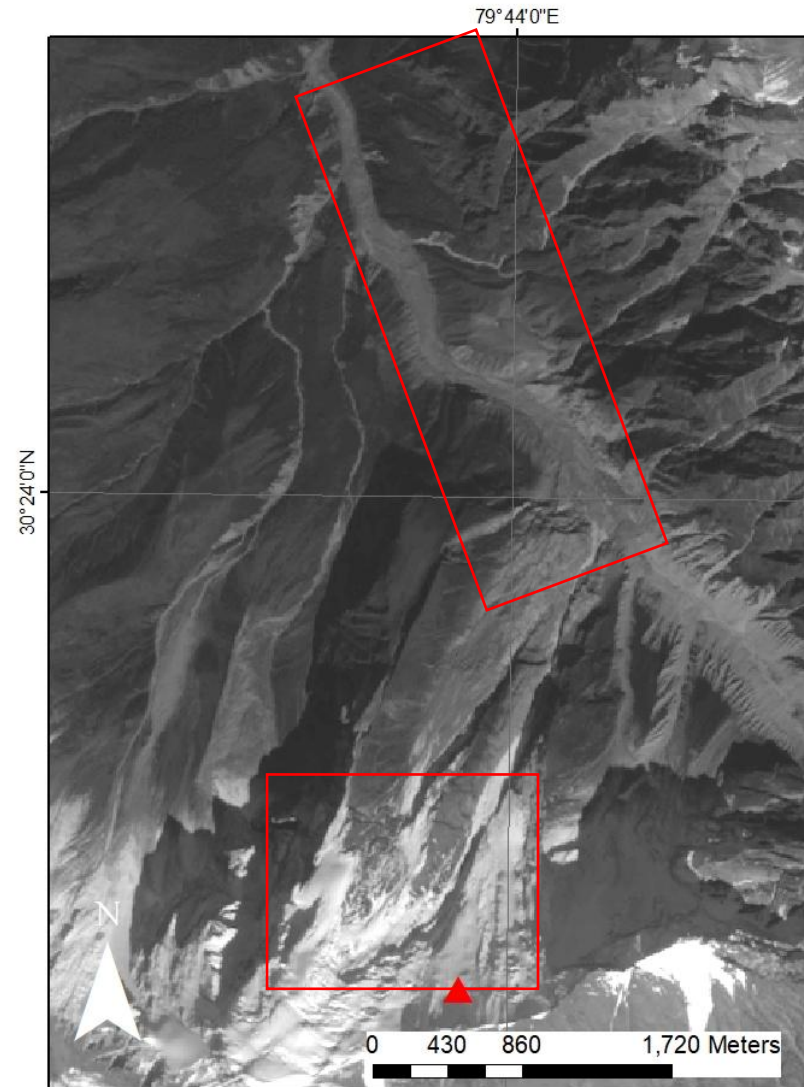
Transition into debris flow



Climate change? Bad luck?

Previous ice avalanche in 2016

Ice/debris deposited in valley below

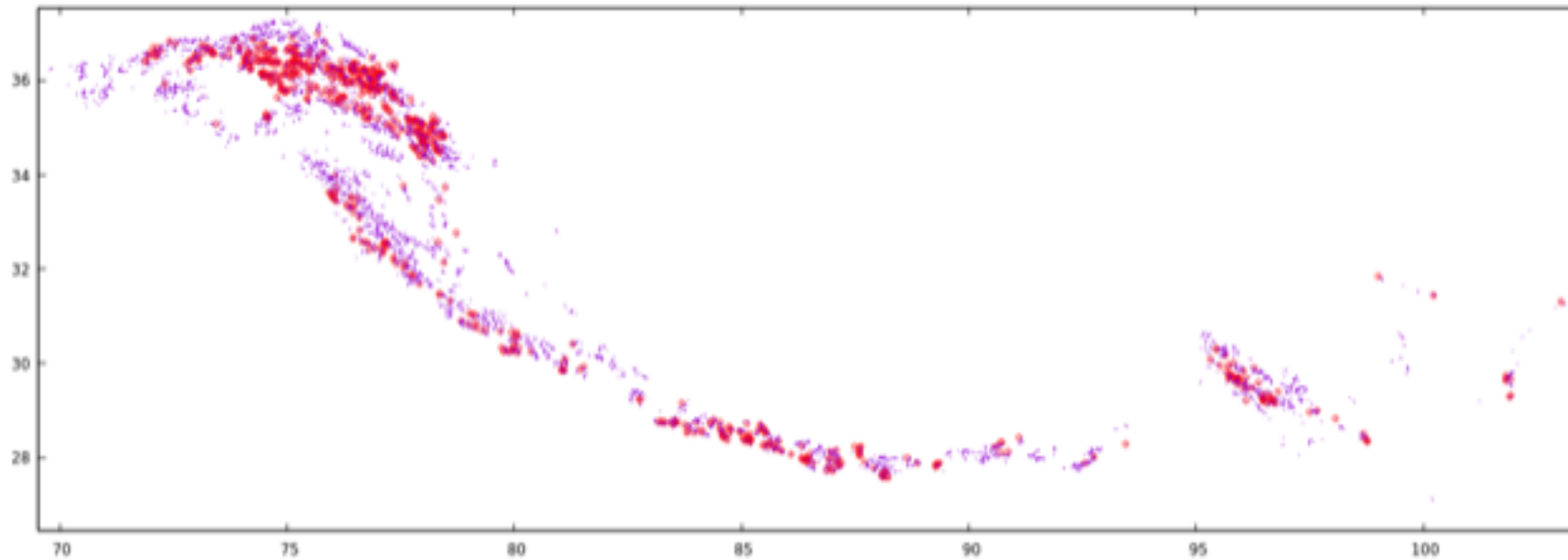




Is scaling this up possible?

Hanging glaciers in the Hindukush – Himalaya

- steep slope and small size

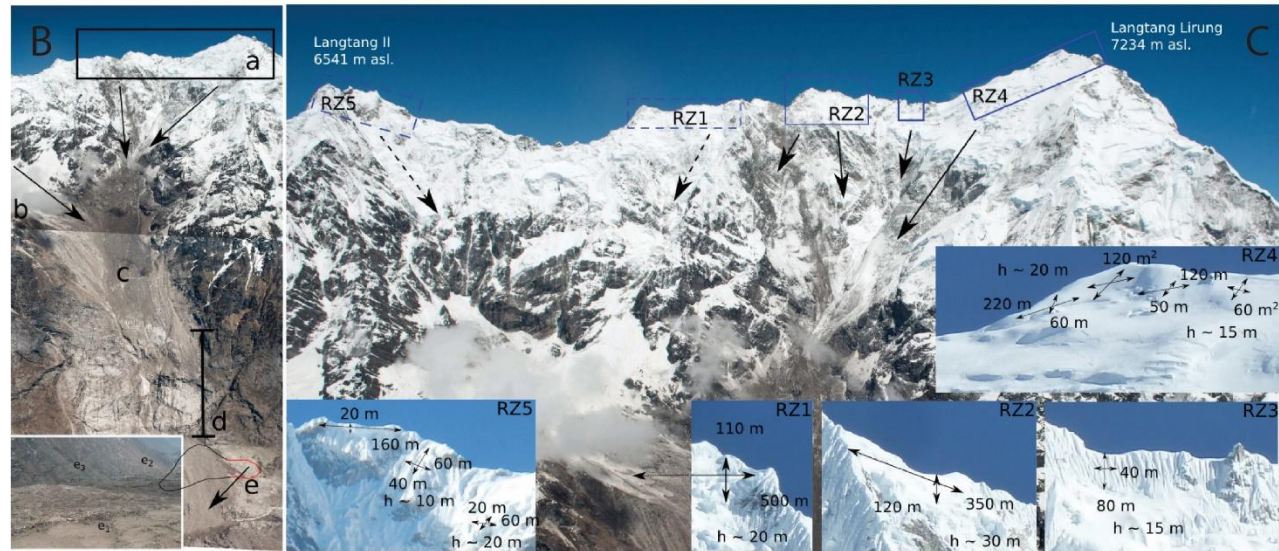
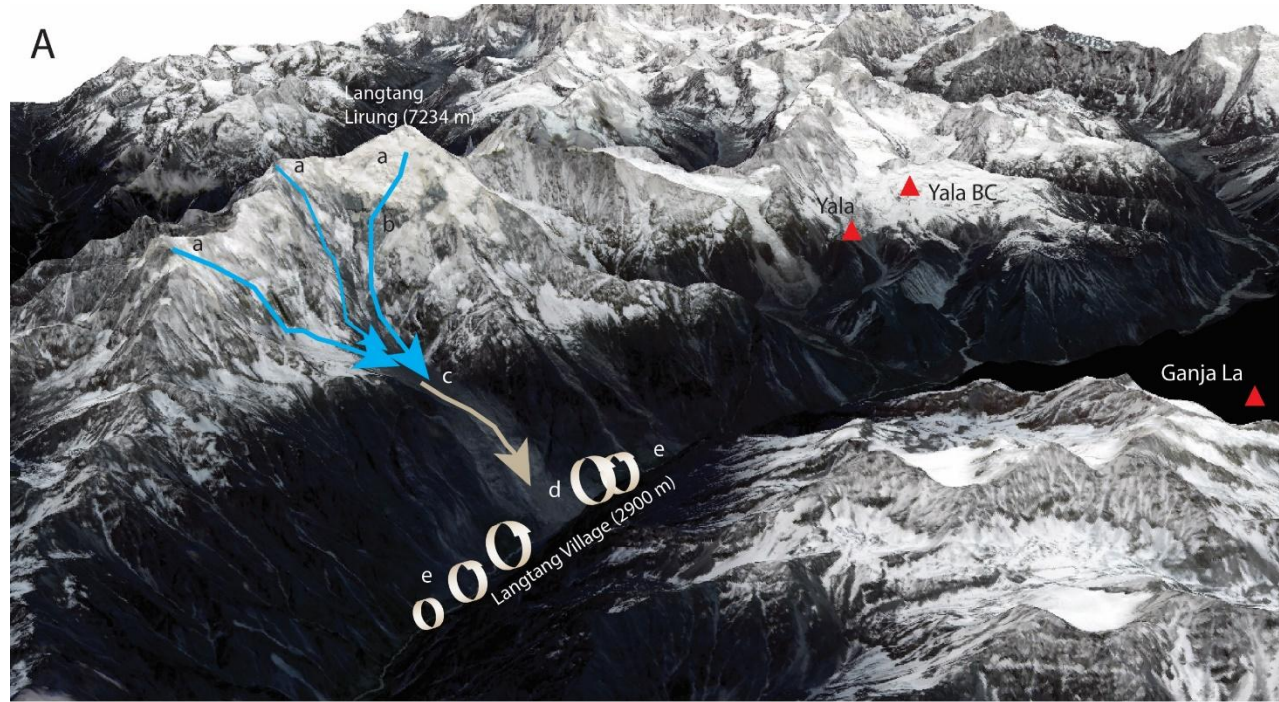


Results – co-seismic events

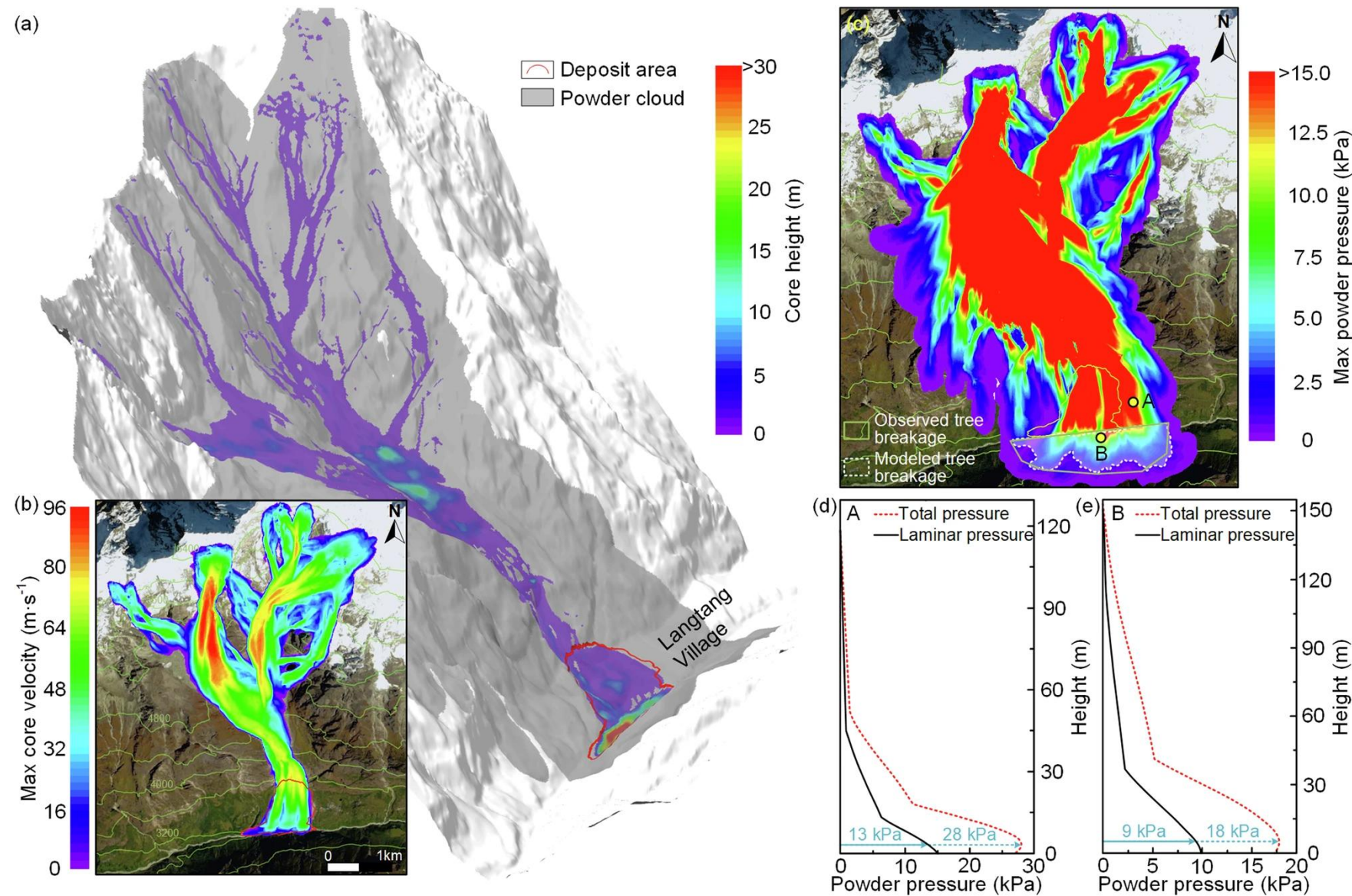
- Gorkha earthquake 2015 in Nepal
- >4000 co-seismic landslides
- Ice-break offs that caused deposits of up to 100 m depth
- Multi-hazard (ice break off + avalanche + debris flow)

Kargel et al., 2016

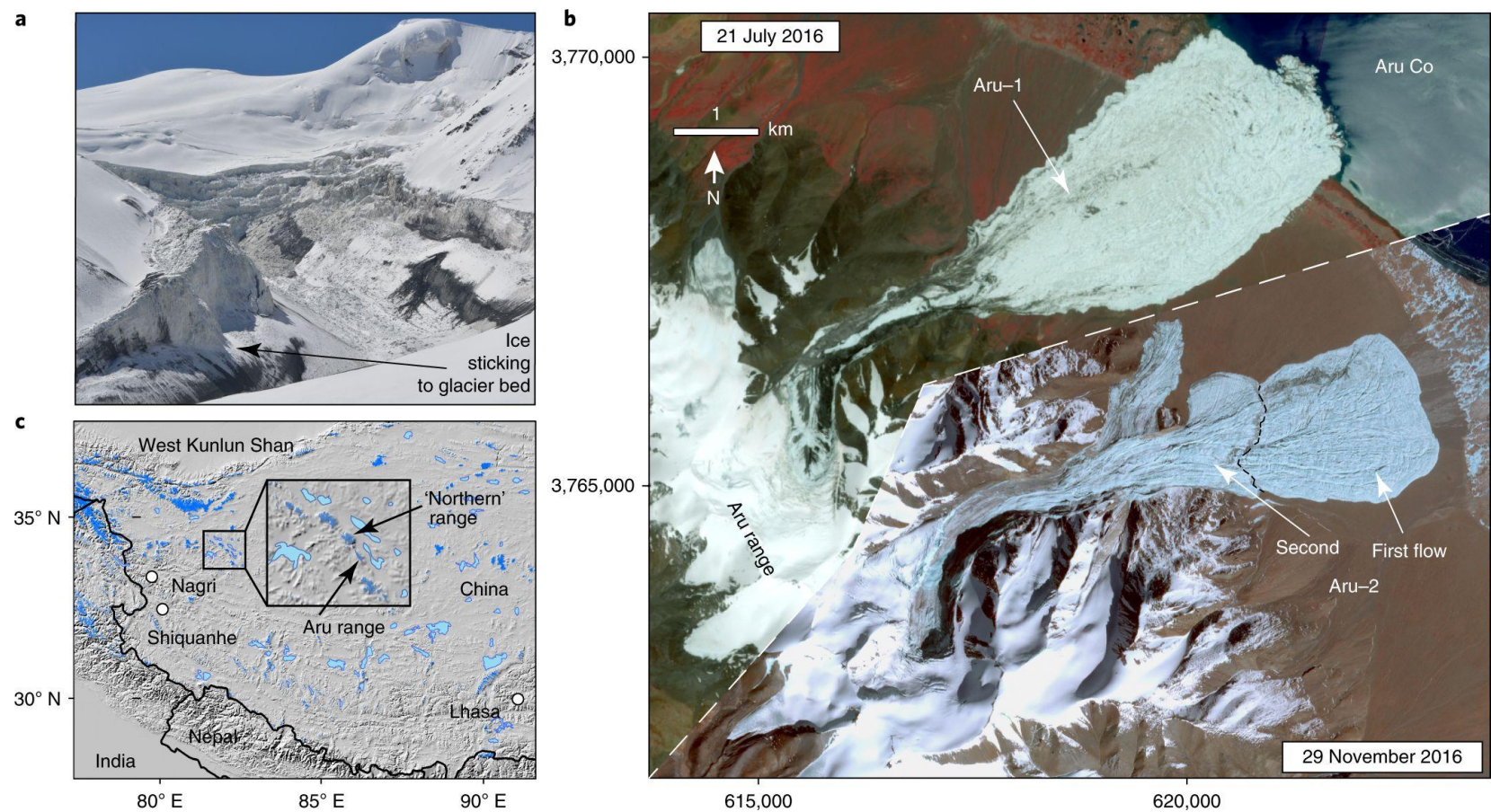




Results – co-seismic events



Results – Internal glacier collapse



- Regional temperature increase leads to internal 'melt lakes'
- Once pressure becomes too high and ice unstable, the glacier 'fails'
- So far singular events (2) but with catastrophic consequences (>125 deaths in Russia during the Kolka slide in 2002)

Kaab et al., 2018

Gilbert et al., 2018

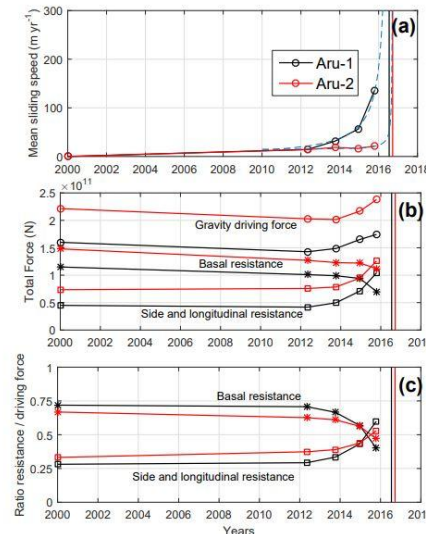
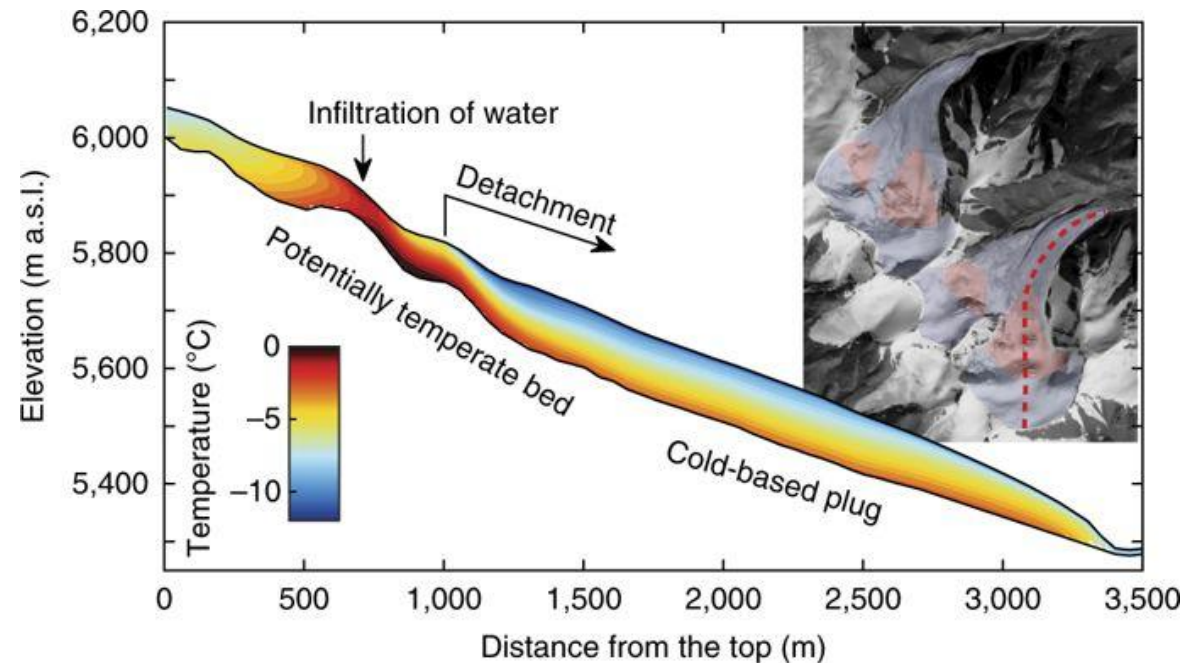


Figure 9. (a) Mean sliding speed of the detachment zone for Aru-1 (black) and Aru-2 (red) glaciers. The dashed blue lines show predicted speed following an empirical law of slope failure (Voight, 1990). (b) Force balance acting on the detachment of Aru-1 (black) and Aru-2 (red). (c) Ratio of resisting force over driving force for Aru-1 (black) and Aru-2 (red). Vertical lines show collapse dates in the three panels. (d) The Aru-2 glacier detachment zone on 4 October 2016 (Picture from T. Yao). The side resistance computed in the force balance analysis arises from lateral shearing on the right side of the detachment (black arrow).

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ice break offs (+ resulting events) / collapsing glaciers

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destabilization of frozen soils/slopes

... what appears when the ice is gone?

Glacier length variations surges

Glacier tongues extend within days to weeks by 100s of meters

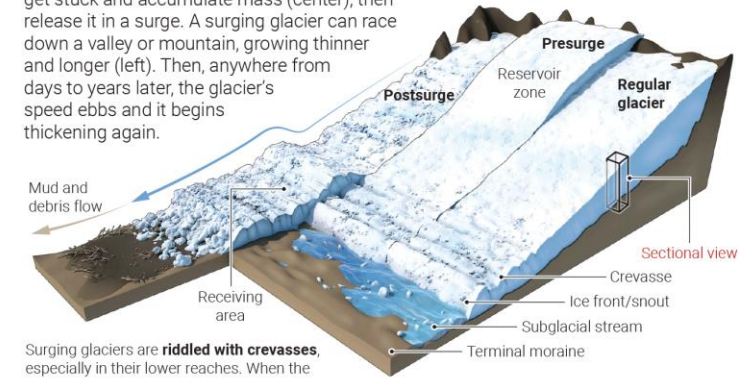
Decadal return period

No net mass gain or loss

Unclear why these events happen – no apparent link to changing climate

A glacier unleashed

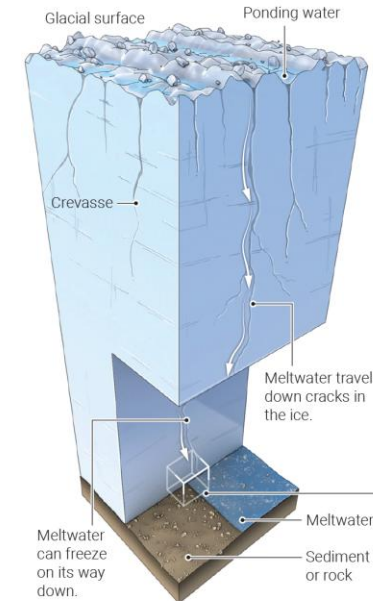
Glaciers gain mass in their upper reaches, where snowfall is heavier, and lose it at their snouts, where the ice breaks up and melts (right). Most glaciers flow steadily, but some get stuck and accumulate mass (center), then release it in a surge. A surging glacier can race down a valley or mountain, growing thinner and longer (left). Then, anywhere from days to years later, the glacier's speed ebbs and it begins thickening again.



Surging glaciers are riddled with crevasses, especially in their lower reaches. When the surge ends, meltwater that built up under the glacier before the surge may sweep mud and debris from its snout.

Trickling down

Meltwater plays a key role in triggering surges. Pooling on the glacier's surface, it can seep down into crevasses. There it can refreeze, releasing heat that softens the ice; it can also pool at the base of the ice.

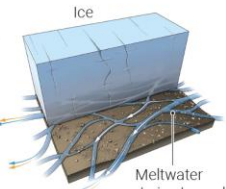


CREDITS: (GRAPHIC) C. BICKEL/SCIENCE

Steady state

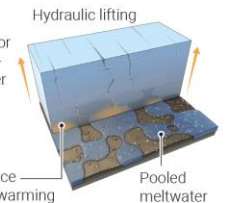
In a "normal" glacier, meltwater drains efficiently from its base, carrying away heat and leaving the ice anchored to its bed.

Steady meltwater flow under glacier releases heat and pressure.



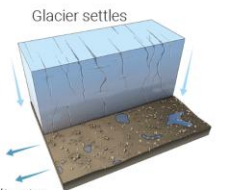
Buildup to a surge

If drainage is poor or melting accelerates, meltwater can accumulate under a glacier, warming the ice and lifting it off the ground.



Aftermath

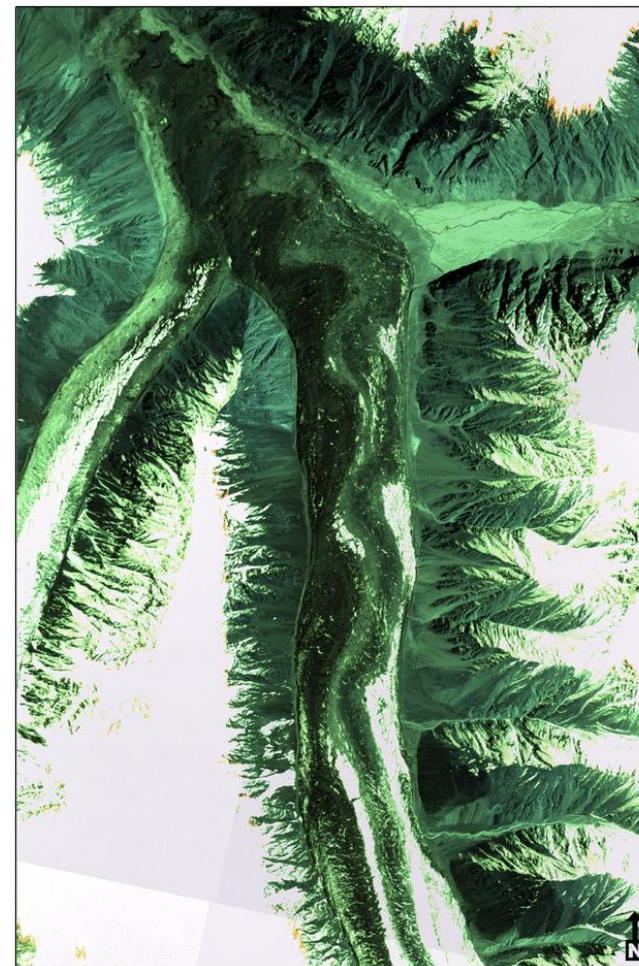
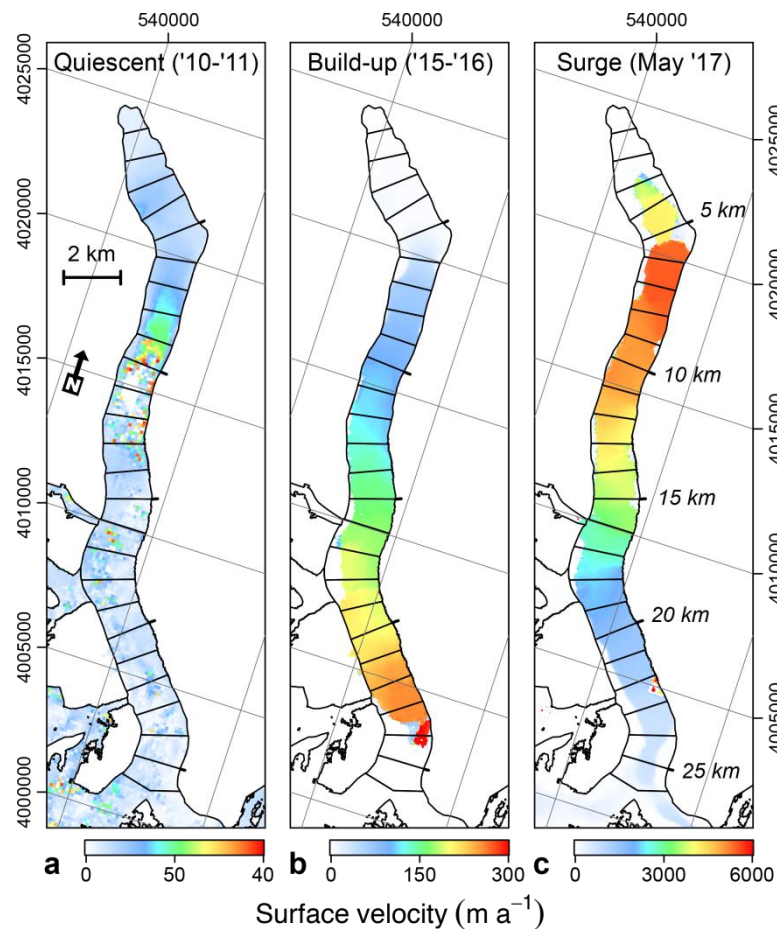
Once the surge releases the meltwater, the glacier subsides onto its bed, and the cycle begins again.



Majority of meltwater expelled during surge.

Surge in Karakoram 2017

Glacier velocities from remote sensing



Steiner et al., 2018, Cryosphere

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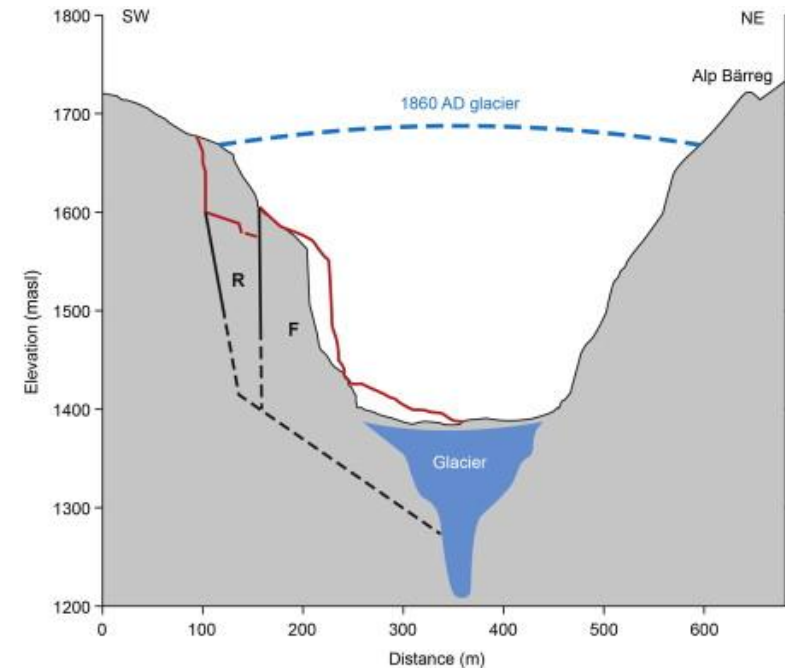
Destabilization of frozen soil

Increased rock slope failure in post-glaciated environments

Dependent on geology

Freeze-thaw within rock cracks

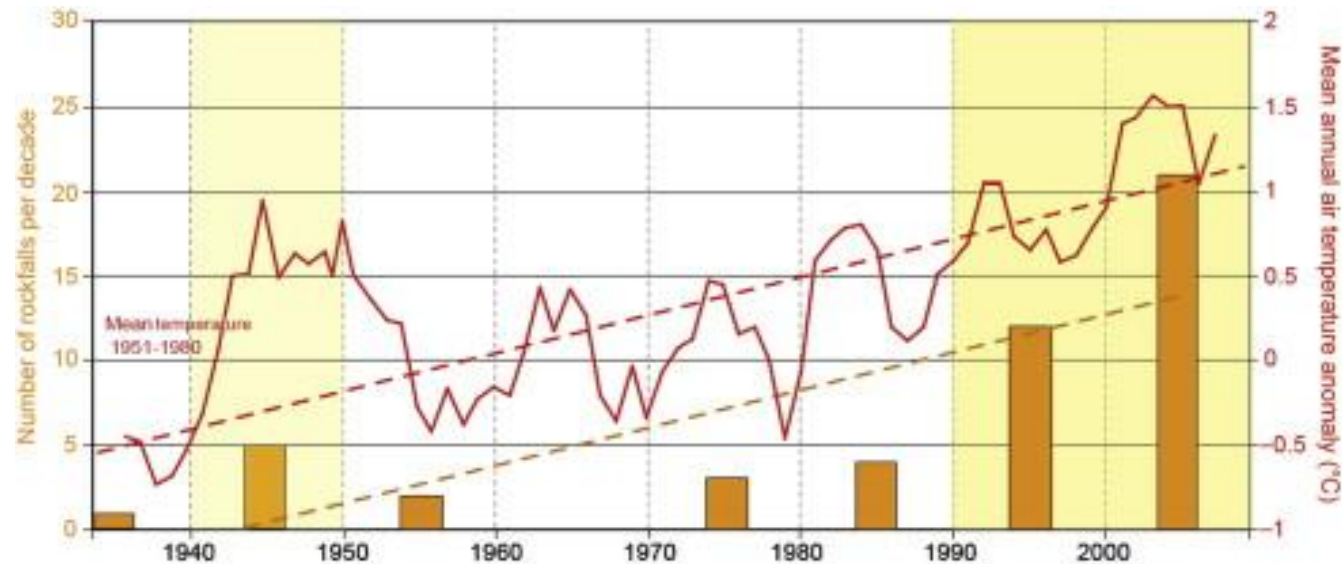
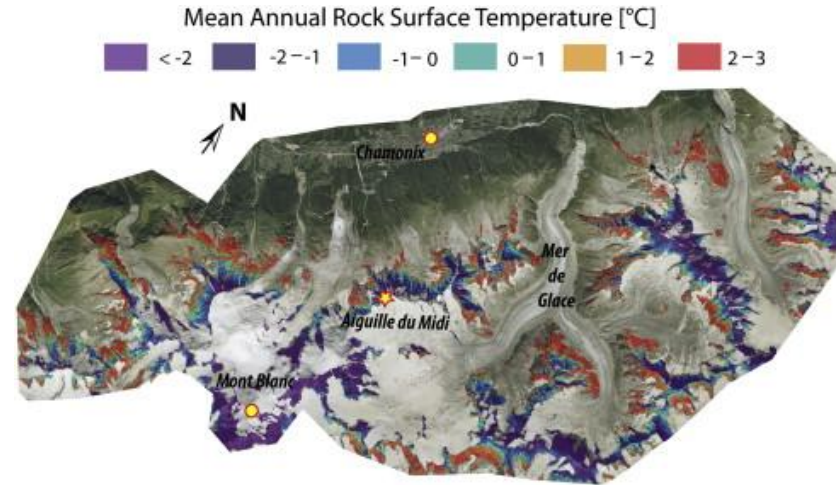
Isostatic rebound after deglaciation



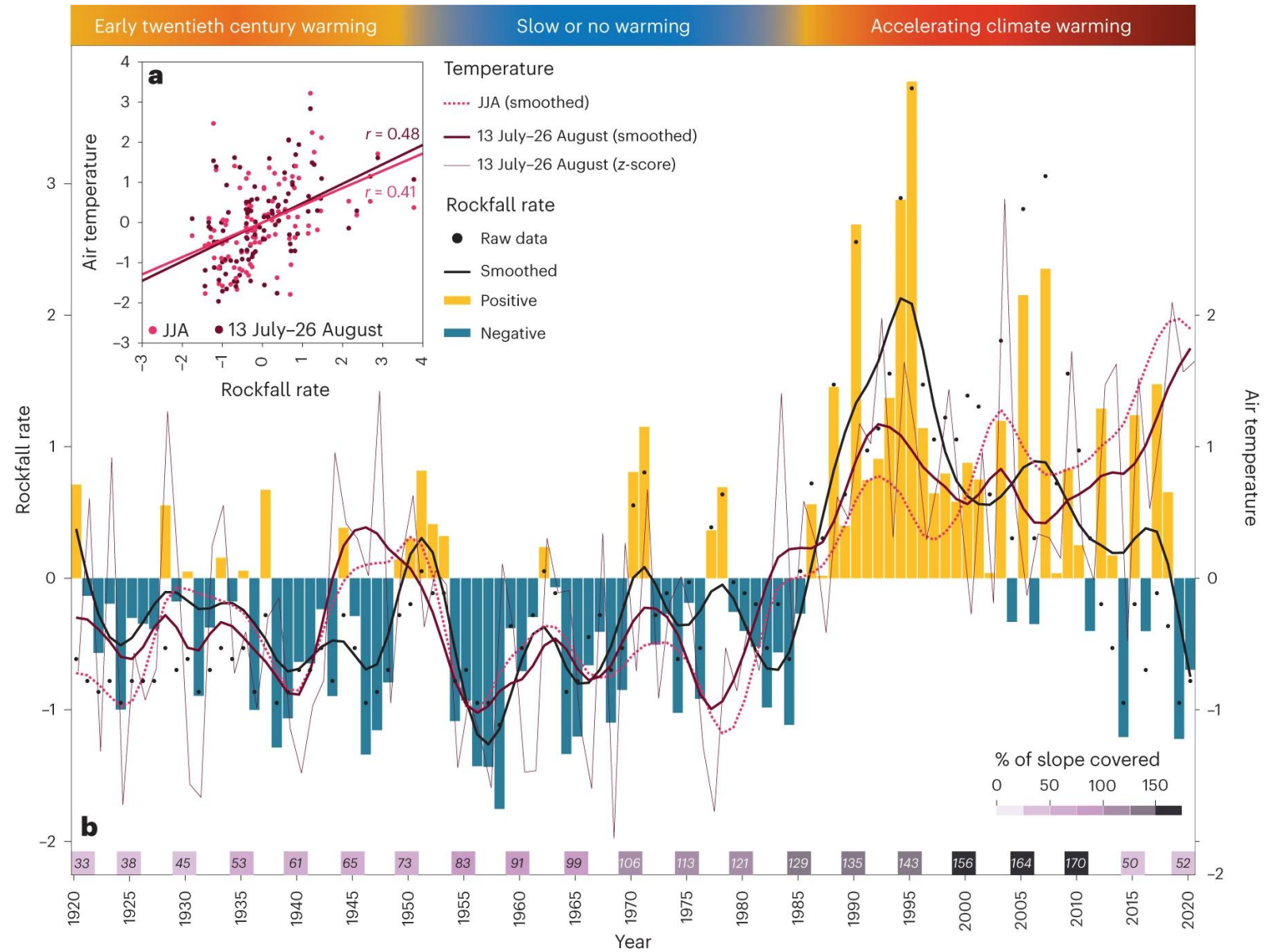
Deline et al., 2015

Destabilization of frozen soil

- Likely direct link with increasing temperatures

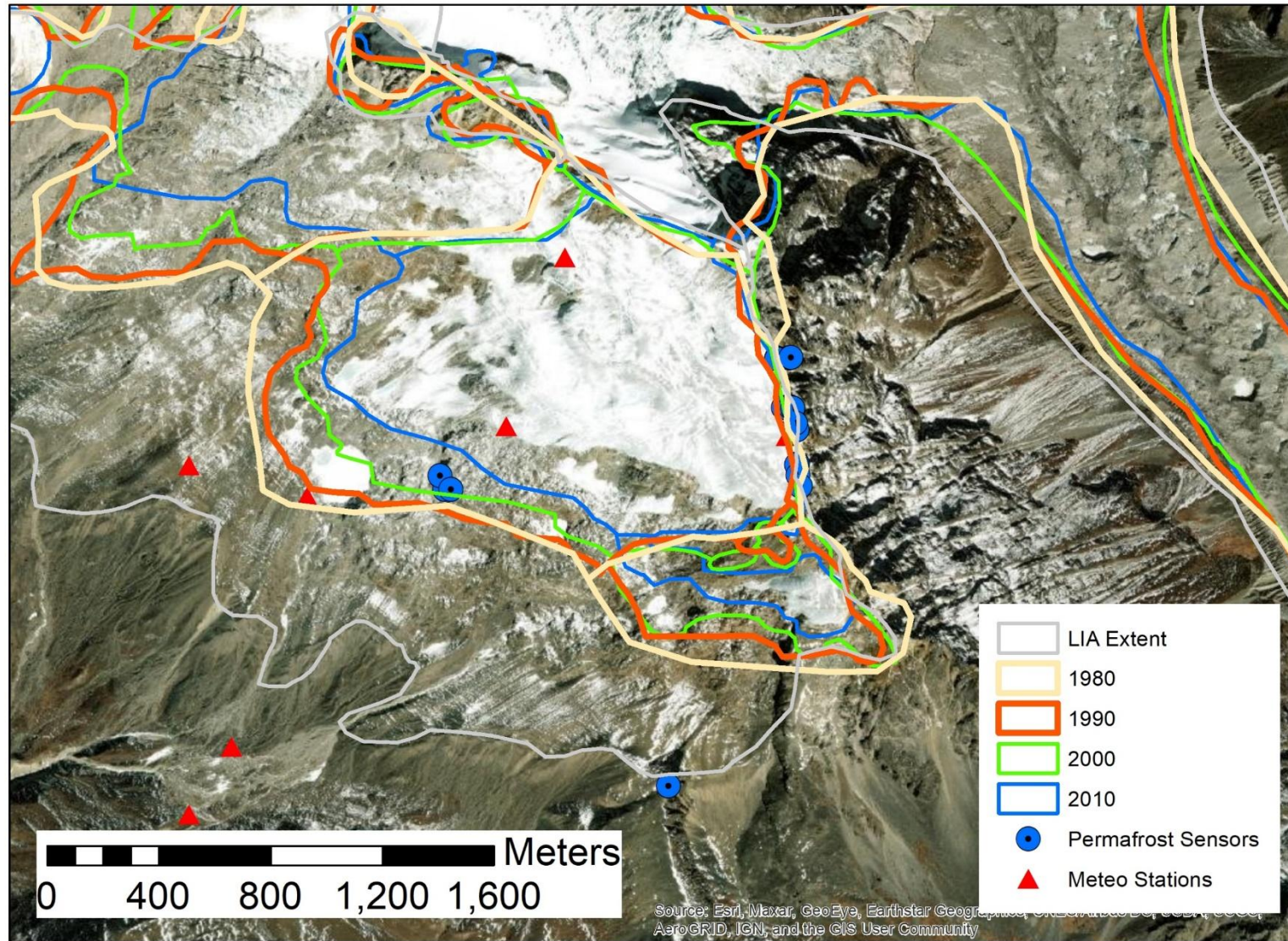


Deline et al., 2015



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Effects downstream – debris flows



Destabilization of frozen soil



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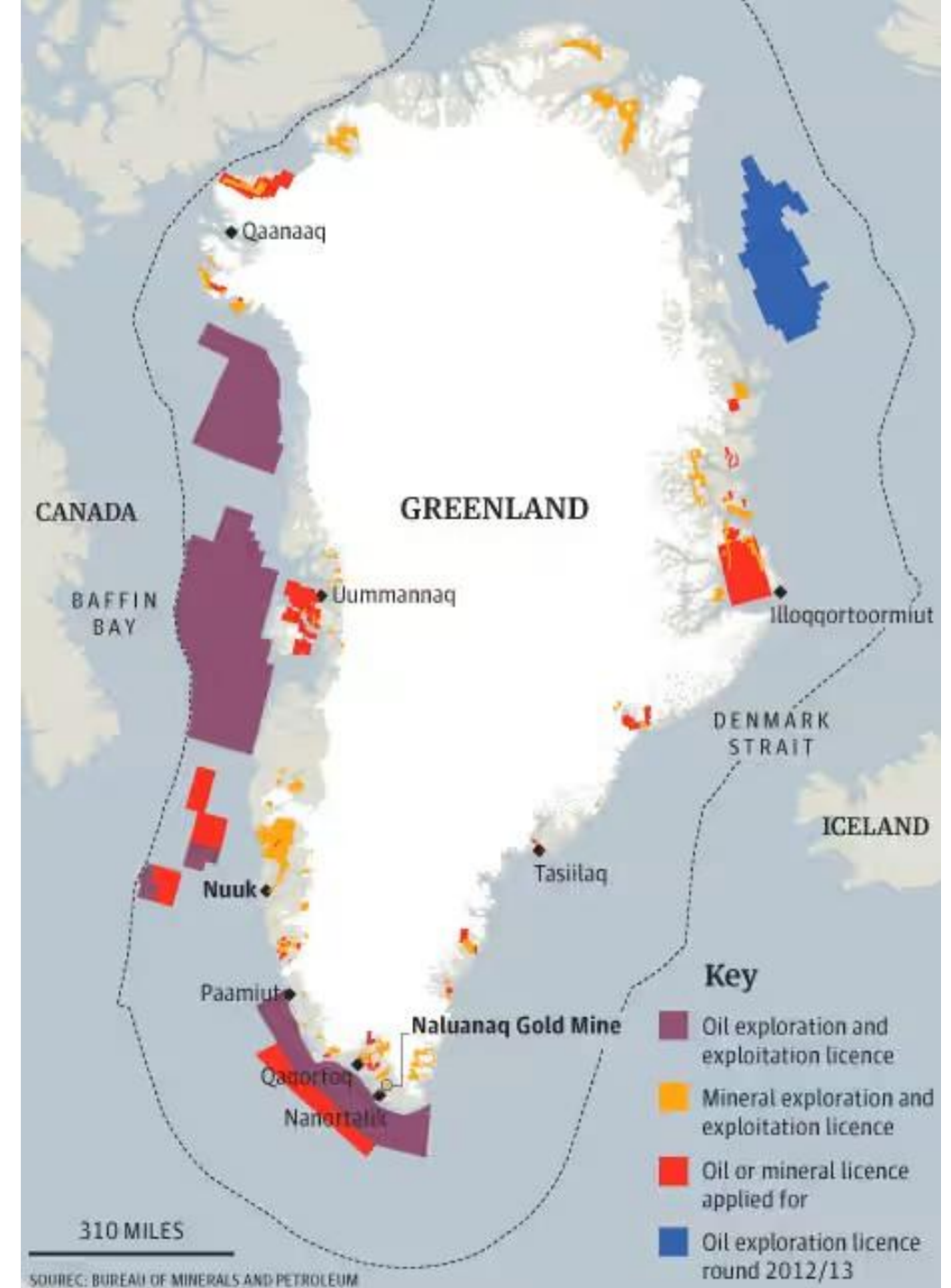
Global ice and exposure to hazards

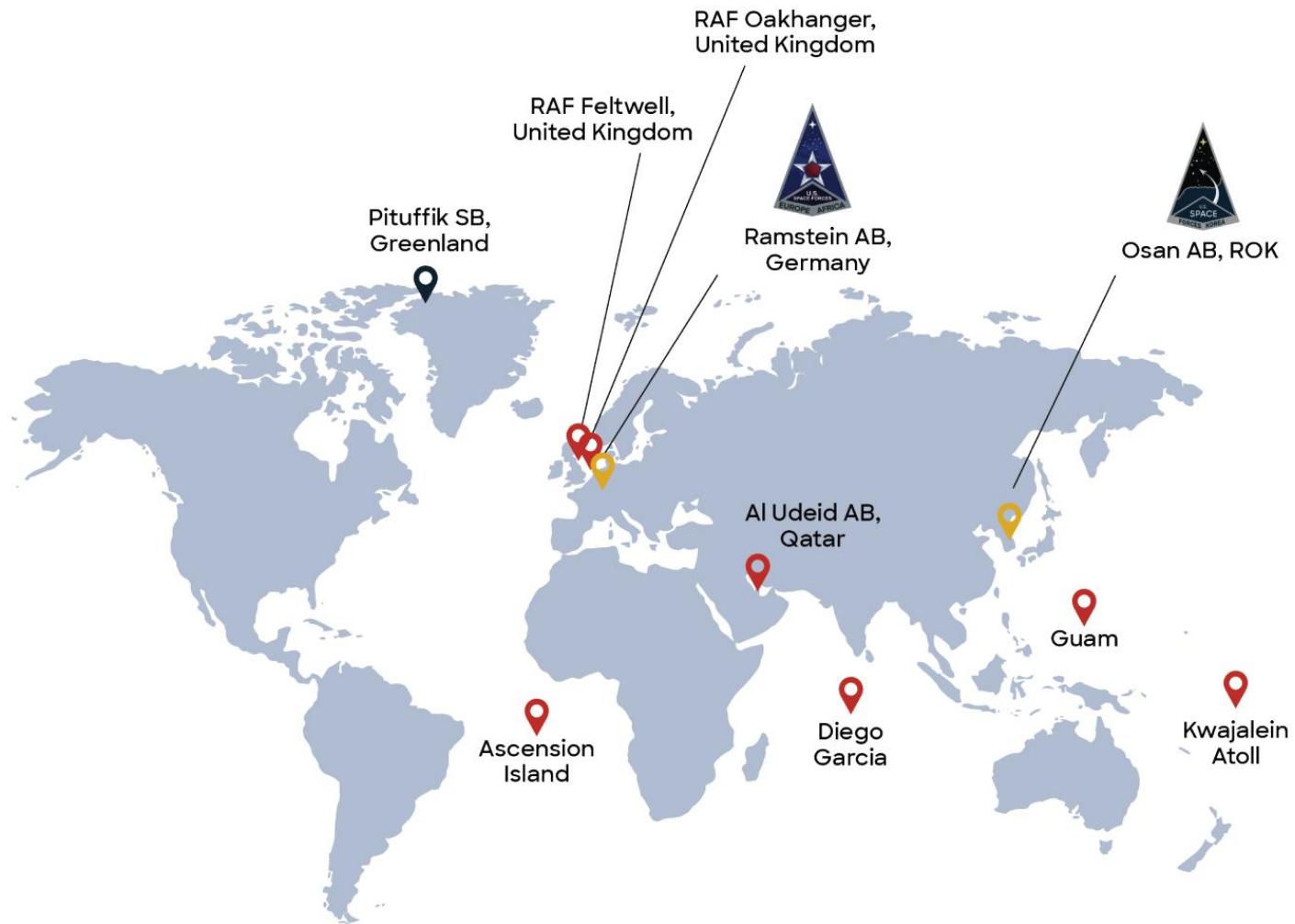
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The global politics of 'glacial risks' - excursion to Greenland

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LEGEND

- | | |
|--|---|
|  U.S. Space Force HQ |  U.S. Space Force Delta HQ / DRU |
|  U.S. Space Force FLDCOM HQ |  U.S. Space Force Base (SFB) / Station (SFS) |
|  U.S. Space Force Component FLDCOM HQ |  U.S. Space Force Sites |



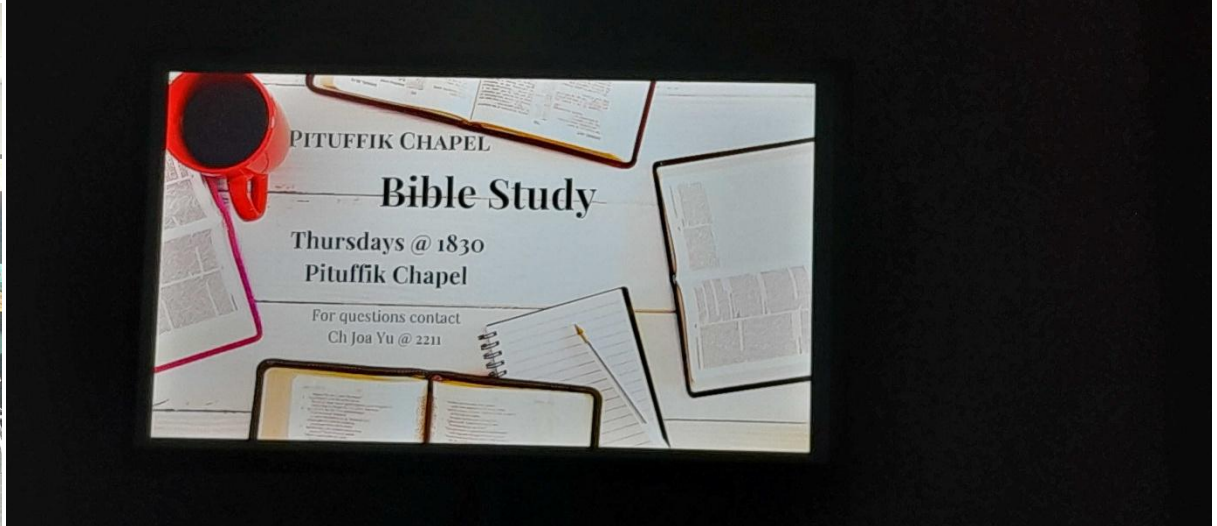
Pituffik Airbase (formerly Thule Airbase)

Built to current extent during the cold war

Serves until today as important satellite downlink and intercontinental rocket early warning site

Displaced local population







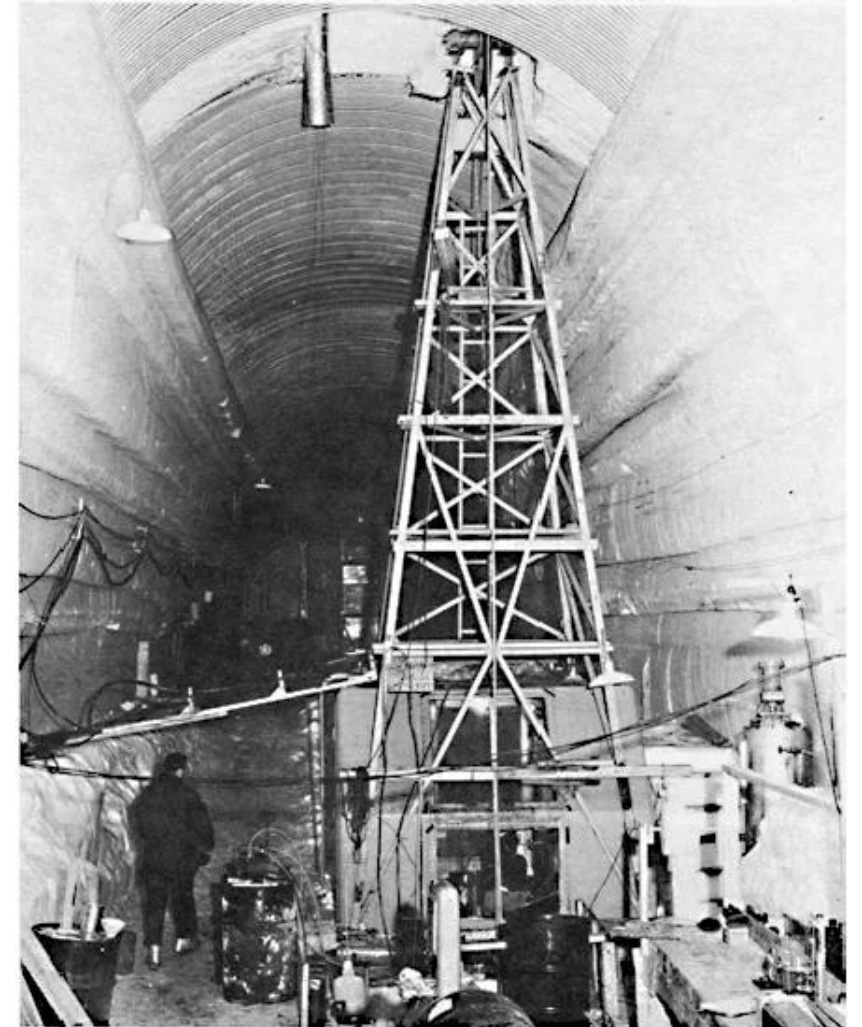




Regulations and oversight in an extreme environment?



1968 Thule Air Base B-52 crash



Camp Century (1950s/1960s)

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The global politics of 'glacial risks' - excursion to Greenland

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Monitoring and risk mitigation

Mapping and modelling

- The basics not any different from elsewhere
- Extra challenge:
 - Lack of data
 - Difficult topography for satellite imagery as well as numerical modelling
 - Access to affected populations difficult (including language/cultural barriers)

Hazard Mapping

Remote sensing

- Maps of potential instable areas
- Identification of weak moraines/ice cores/rapid surface changes on glaciers
- Slope maps + exposure to solar radiation (freeze thaw cycle)
- Exposure maps

Field based

- Runout lengths
- Damage classification (pressure, inundation)
- Interviews

Impact - observed



<https://hirisk.org/risk-assessments/>

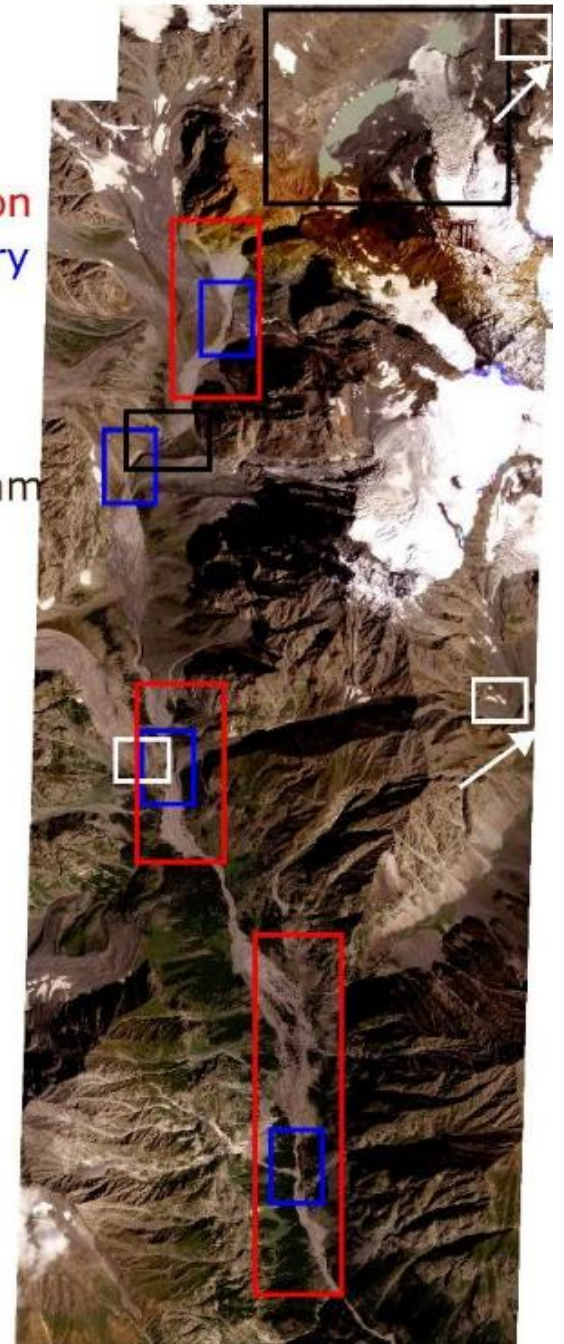
drained lake

significant deposition
significant secondary
erosion

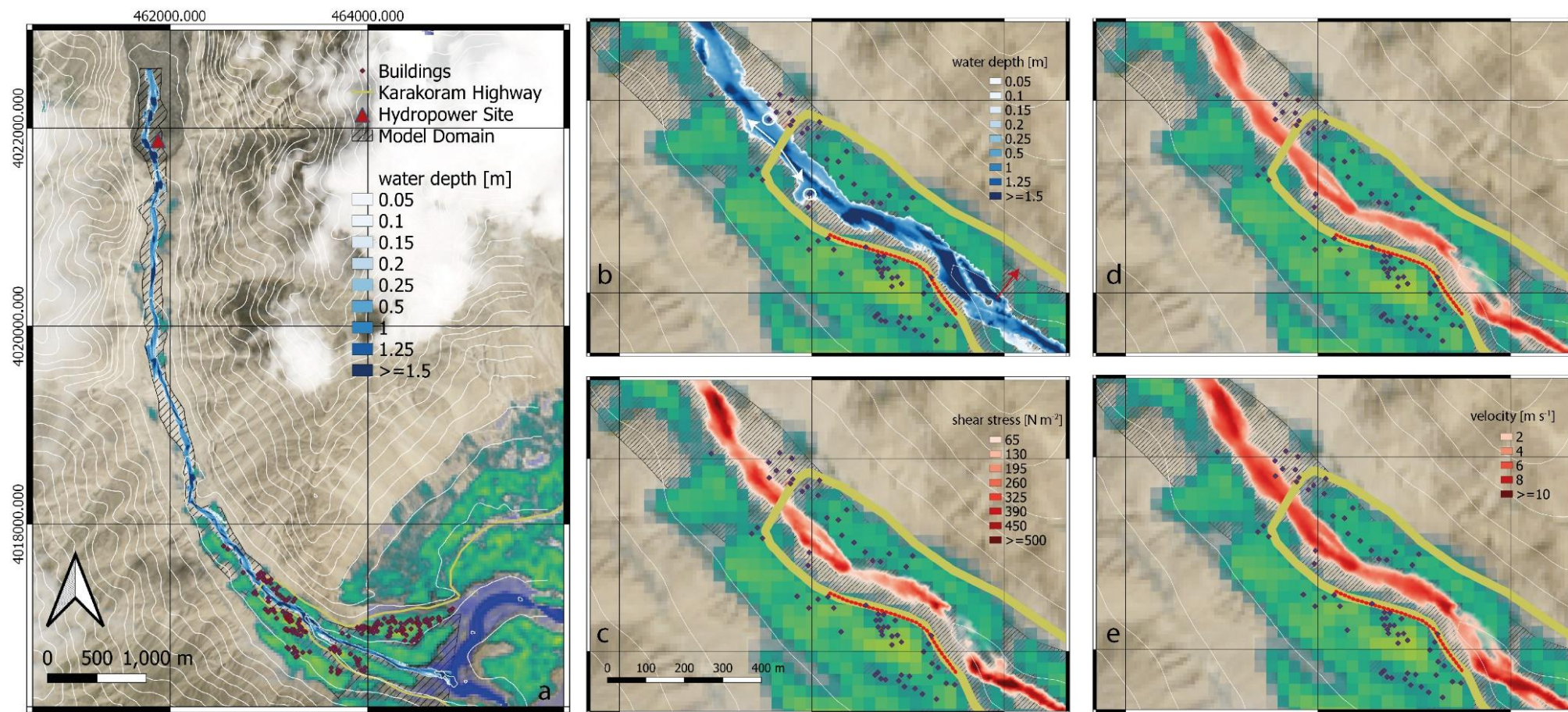
Shazore lake and
likely temporary dam

additional glacial
lakes

0 750 1,500 m



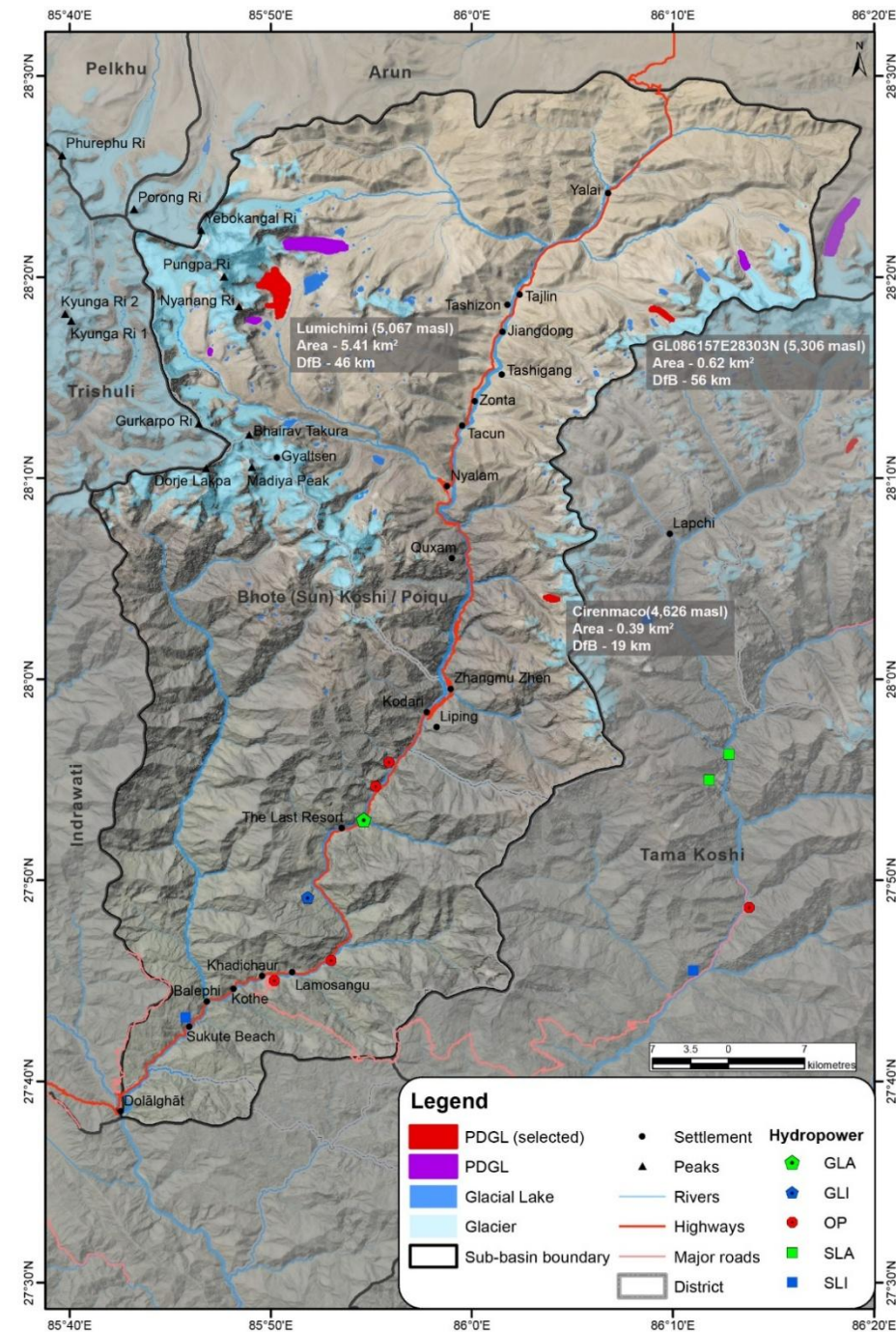
Impact - modelled



Muhammad et al., 2021

Challenge across borders

- Transboundary multi-hazard and risk assessment
- China-Nepal border



Local and Indigenous knowledge

- Focus Group Workshops
- Interviews
- Awareness Raising



Shimshal, Pakistan



Melamchi, Nepal

Local and Indigenous knowledge

- How do we turn local knowledge into (western) scientific data?
- To whom do glaciers matter, how and when?



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Monitoring and risk mitigation

Monitoring and hazard mitigation

Remote Sensing

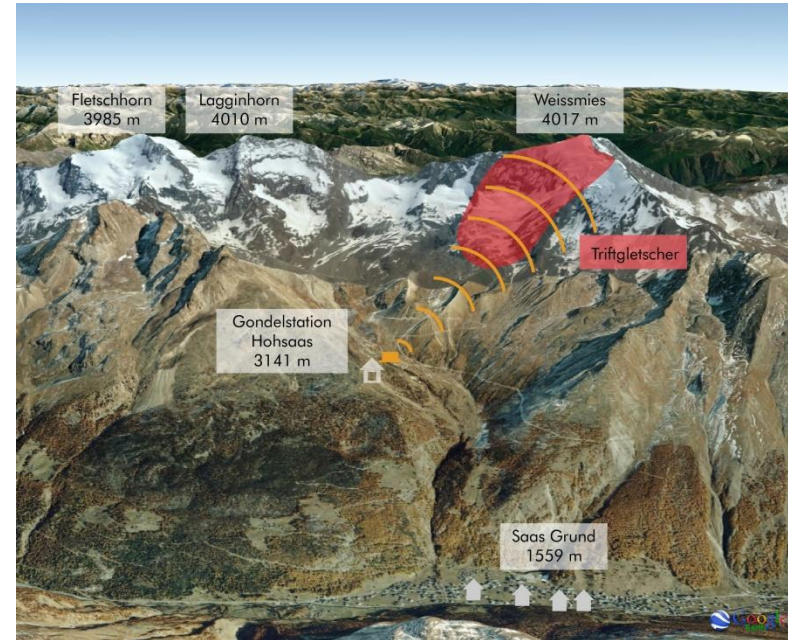
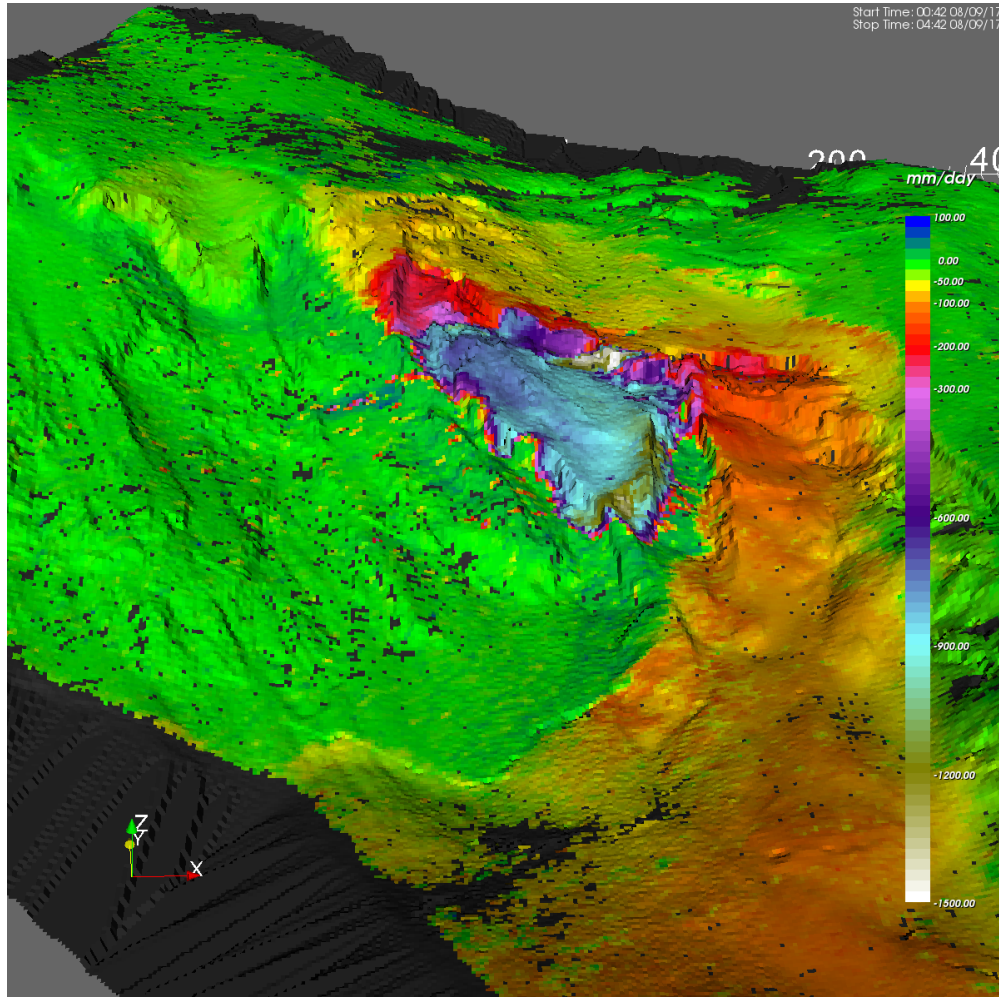
- new near real-time products for observations (Planet, Sentinel, new Radar satellites ...)
- Temperature/Precipitation products

Field based

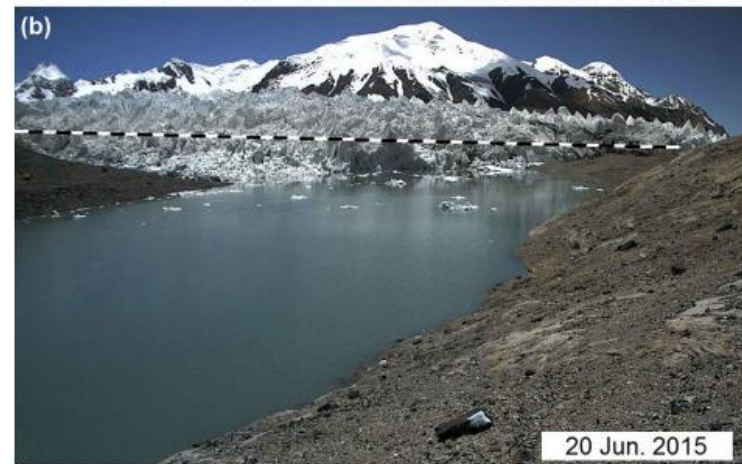
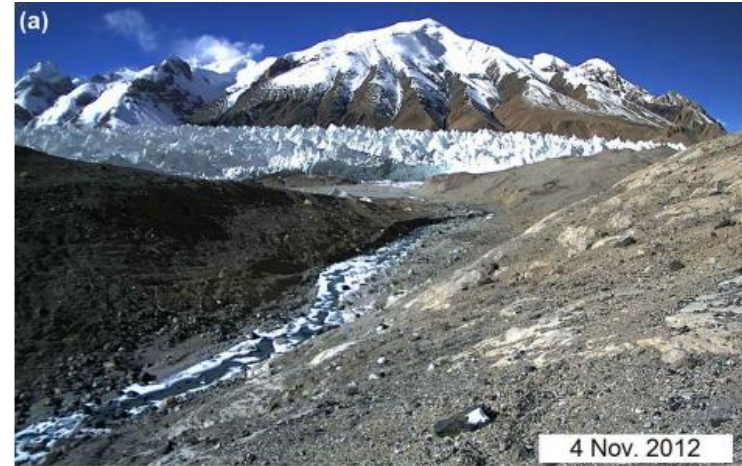
- climate and discharge monitoring
- radar monitoring

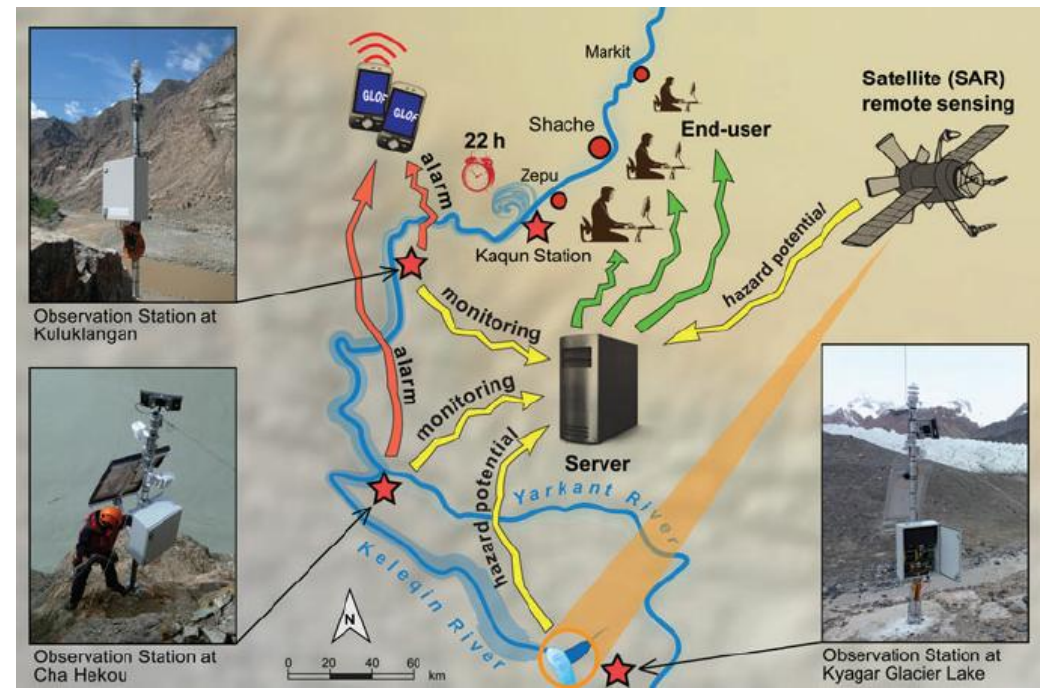
Local knowledge?

Monitoring ice falls

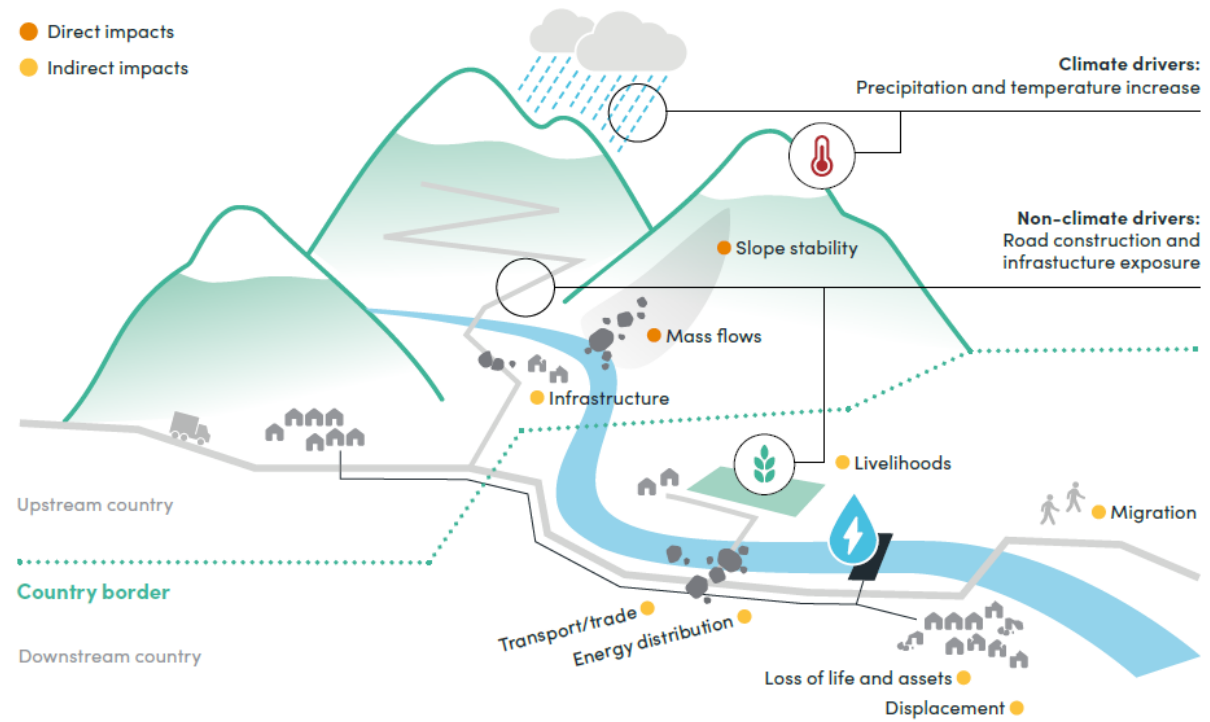


Monitoring a glacier lake





Risk



Steiner et al., 2023

Upstream – downstream linkage

Cascading hazards and risks (*knock on effects*)

Transboundary risk

Local challenges – Alps vs Himalaya

- Similar glacial hazards
- Population in relatively remote areas directly exposed
- Expensive infrastructure (Nepal/Pakistan: trade to China; India: connection to tourism areas; Switzerland: everything is expensive ...)
- (Geo-)Politics? Willingness/ability to pay? Institutional development?



Grindelwald Switzerland

Glacial lake formed repeatedly

Concerns about possible downstream damages

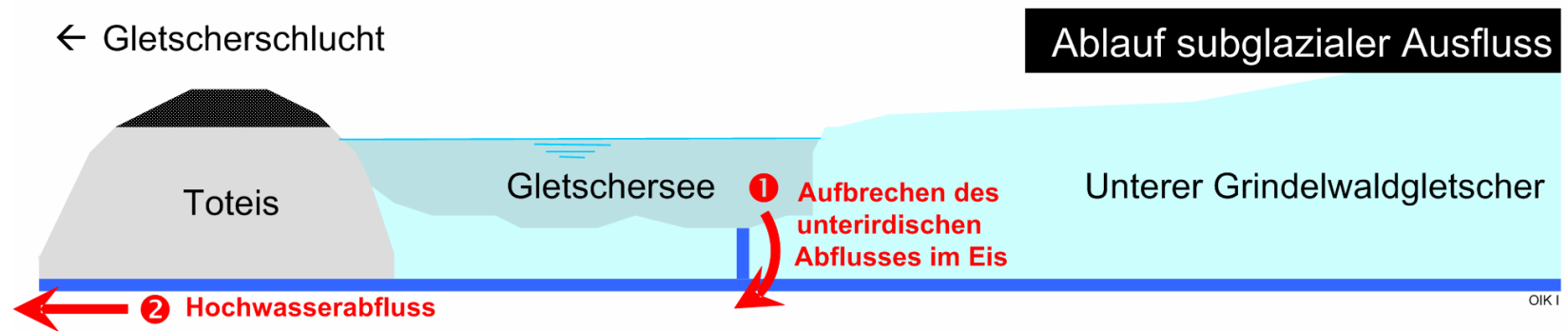
Classical compound event

Who decides? Who pays?

Grindelwald Switzerland



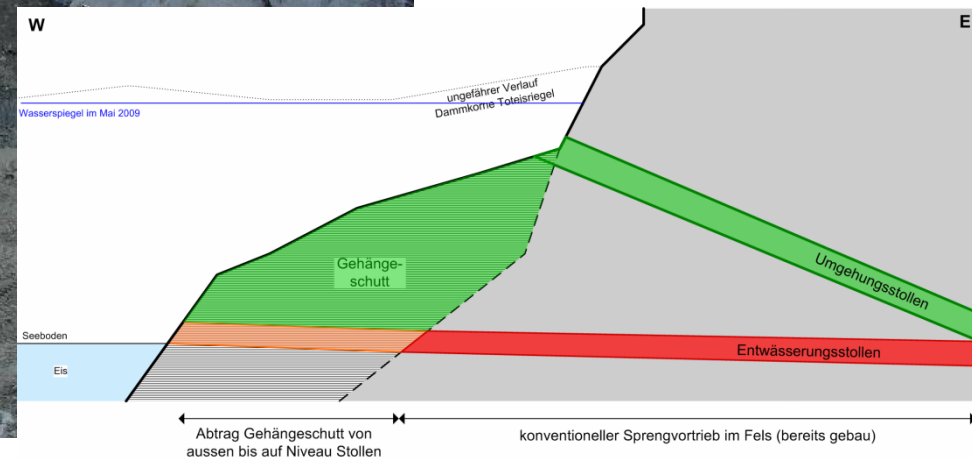
Formation of a glacier lake



Slumping moraines and rockfall



Decision to build a tunnel ...



... and then the lake emptied naturally



Shimshal glacier lake (Pakistan)

Lake forms as a result of a glacier surge

Damages to local roads and agriculture

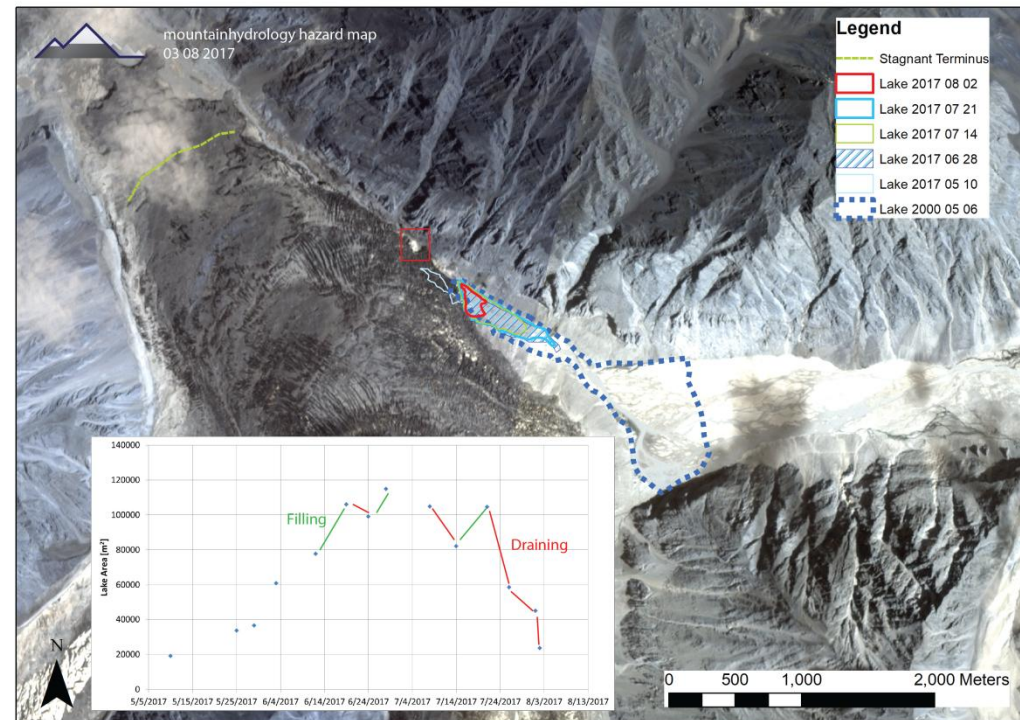
Politically instable

Difficult to access (no telephone, bad roads)

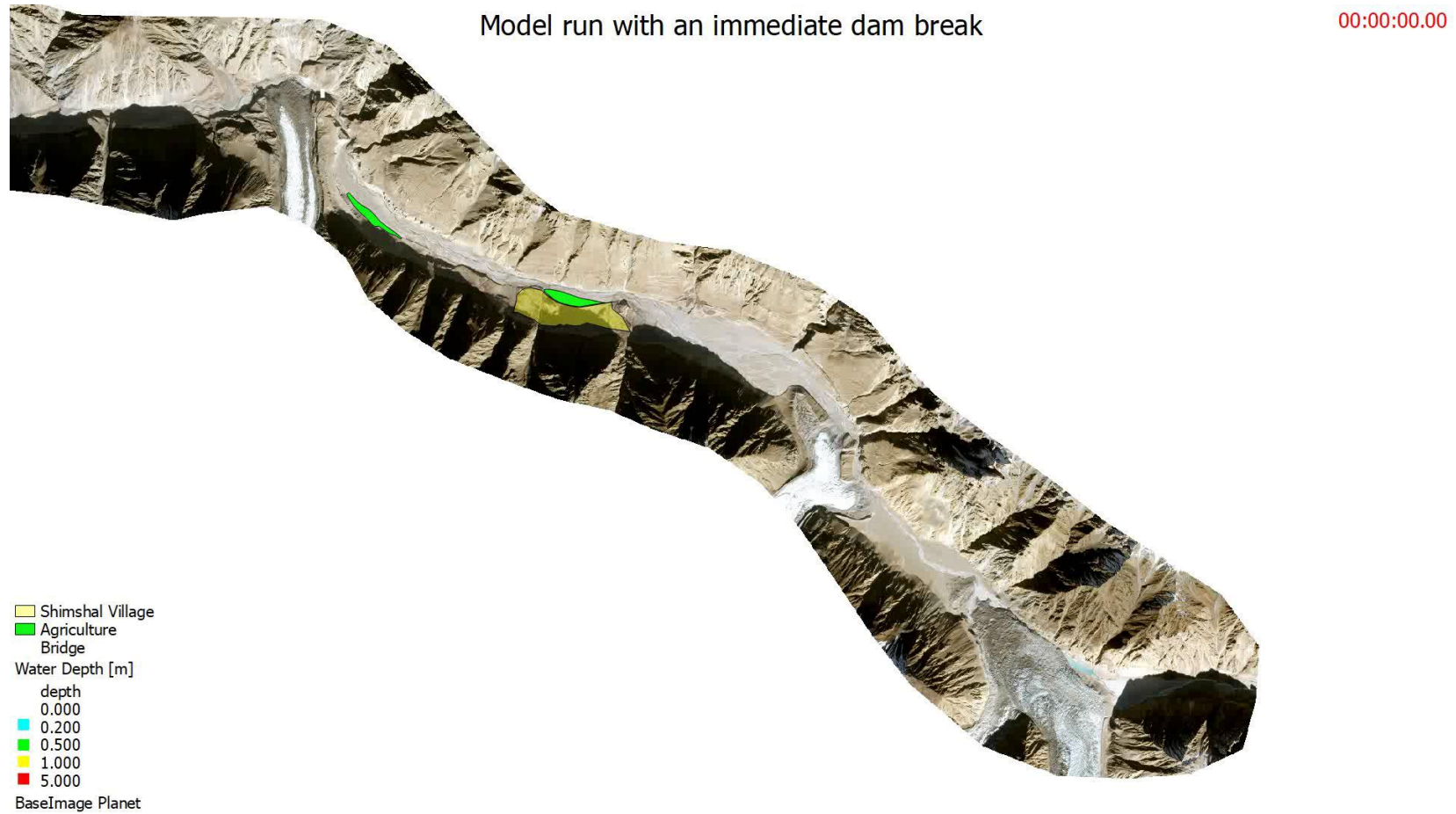
Scenarios

Volume = $f(\text{Area})$

Discharge = $f(\text{Volume}, \text{dam break scenario})$



Hydrodynamic modelling



Institutional challenge

